Utility-Gated Collective Semantic Memory: Formation, Admission, and Long-Horizon Coordination

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Abstract

This paper studies long-term collective memory as a *formation-and-admission process* rather than a shared archive, in two connected halves. The first establishes the minimal mechanics. Treating collective memory as a monotone resource fails under semantic target scarcity: fast sharing improves target materialization but reduces completion because several agents act on the same records. A minimal collective semantic memory (CSM) that separates source-aware merge, explicit trust, and staleness-gated consumption occupies a distinct, better region of the materialization–completion plane, and the benefit comes from selective admissibility, not communication volume. Removing the shared-frame assumption through semantic place fingerprints preserves this behaviour across private frames; an order-theoretic account (deferred to the appendix) interprets the merged memory as a coherent gluing of local views whose consensus-spanning is a measured, trust–cadence-dependent property. The second half studies how records are *formed* and *admitted*. Retention is a two-axis decision — counterfactual marginal utility (worth retaining) times structural analogy (how to integrate) — realised by five operations (merge, exception, new schema, repeat, drop) over an immutable episodic store. The formation policy is learnable (a contextual bandit beats the hand-designed rule) and a utility estimator matches exact counterfactual replay on held-out worlds (Spearman 0.989), so at deployment the runtime requires no online counterfactual replay — after calibration on replay-accessible training environments. On long horizons, unregulated exchange is harmful; temporal decay of foreign testimony is necessary but insufficient in the evaluated drift setting — and a clock-alternatives study shows the operative signal is receiver-relative (age since own corroboration beats a world clock, while ordering-only Lamport/vector clocks are worse than no clock); receiver-side admission — validating foreign testimony on the recipient's own visit — reverses the sign: mean cumulative cost falls below independent memory on both roles (fragile 118 vs. 148, robust 134 vs. 171; $\approx 20\%$ relative), and a staged component ladder attributes the reversal to quarantine plus admission, with reservations adding duplicate hygiene at no cost. An integrated single runtime combines formation, quarantine, admission, private-frame alignment, route execution, reservation, and rollback; at equal memory and communication budgets it dominates raw history and independent memory, survives perception noise and adversarial provenance, and transfers to synthetic-continuous and VMAS substrates with fail-closed safety. Controlled LLM experiments show a consolidated Songlines graph resolves freshness and cross-frame identity at a fraction of the raw-context cost; a fixed-budget comparison against five stronger context-management baselines attributes the freshness gain to structured consolidation as such (a programmatic latest-state table suffices), while cross-frame identity remains specific to

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the graph's alignment machinery. The result is an end-to-end research prototype of utility-gated, provenance-aware collective memory within the evaluated controlled substrates; it does not establish open-world symbol grounding, a general category-theoretic theory of agent cognition, broad social-LLM generality, or real-robot validity.

We claim. In the evaluated class of non-stationary, resource-scarce multi-agent settings with heterogeneous roles and potentially stale foreign testimony, long-term collective memory functions not as a shared archive but as a distributed formation-and-admission process. Estimated counterfactual marginal utility governs whether an experience is retained; structural analogy governs whether it is merged, stored as an exception, or promoted to a new schema; and foreign records gain *durable* action authority only after receiver-side admission under explicit merge, trust, staleness, and coordination constraints — while bounded provisional use for transport remains available under the validity contract (companion route-transfer study). Utility gating certifies decision usefulness, not truth: retention reflects a record's measured contribution to task outcome, and the epistemic burden is carried by the provenance and validity contracts, not by the utility estimate.

1 Introduction

A collection of agents does not obtain a useful collective memory merely by placing all observations in one shared store. Our stage-decomposed multi-agent measurements show why: communication improves access to target evidence, but unrestricted or overly frequent sharing can synchronize commitments and reduce task completion. A collective memory must therefore regulate not only what is available, but also which records are admitted into each agent's decision process and how conflicts among local views are resolved.

This paper introduces a minimal collective semantic memory (CSM) based on three explicit mechanisms. The merge rule preserves source and support information when records from different agents are combined. The trust rule assigns source-dependent authority to foreign evidence. The staleness rule limits the duration for which an observation can influence retrieval and materialization. Agents retain private memory and construct local merged views; no central cognitive state or single global interpretation is assumed.

The empirical question is whether these mechanisms produce a distinct operating regime rather than merely another communication frequency. We compare independent memory, a shared bus, a central aggregator, fixed-cadence peer exchange, and the proposed CSM under identical navigation and planning components. We then relax the common-coordinate assumption using semantic place fingerprints and evaluate whether the same physical referents can be aligned from local meaning alone.

The formal analysis serves two purposes. First, it clarifies the conditions under which local place memories can be glued coherently. Each realized memory is represented as an ordered category, and partial alignment maps are constructed from observed correspondences. Their colimit represents the merged structure, while connectedness of the accepted correspondence graph determines whether one consensus component spans the population or whether several sub-consensuses remain. Second, a coalgebraic reading represents agents as continuing processes that emit actions and messages over time, complementing the static algebraic description of memory content. Both readings are interpretations of the implemented system, developed in the appendices; neither is load-bearing for the empirical claims, and neither appears among the contributions.

The contributions of this paper are as follows:

1. A minimal collective semantic memory with explicit merge, trust, and staleness rules, evaluated against four communication topologies and shown to gain from selective admissible consumption rather than communication volume — including semantic place correspondence that removes the shared-frame assumption (formal interpretations deferred to appendices).
2. A two-axis memory-formation rule — counterfactual marginal utility times structural analogy — realized by five operations (merge, exception, new schema, repeat, drop) over an immutable episodic store.

3. A learned formation controller that outperforms the hand-designed rule, and a utility estimator that matches exact counterfactual replay (Spearman 0.989) and requires no oracle at deployment (it is calibrated on replay-accessible training worlds).
4. Quarantine and receiver-side admission that turn long-horizon social exchange from harmful into net-positive (mean cumulative cost 118 vs. 148 and 134 vs. 171 against independent memory, $\approx 20\%$ relative), resolving the social ladder exchange $\rightarrow$ temporal decay $\rightarrow$ admission — with a clock-alternatives study and a staged component ladder attributing the effect.
5. A single integrated equal-budget runtime combining formation, admission, private-frame alignment, route execution, reservation, and rollback, with adversarial-provenance, perception-noise, and continuous/VMAS validation, plus an LLM active-context comparison.

The mathematical account is intended to formalize the implemented merge and alignment operations. It is not used as a decorative claim of novelty, nor does it establish that arbitrary agent societies form a single global mind. The scope remains a controlled family of semantic navigation memories with explicit correspondences and admissibility rules.

Secondary claim. The value of information exchange is produced by the admission and consumption contract that converts foreign testimony into locally validated evidence, not by exchange alone.

Structure of the paper. The paper proceeds in two halves. The first fixes the minimal mechanics — merge, trust, staleness, and private-frame alignment — that make a distributed memory an admissibility gate rather than a shared archive, with the order-theoretic and coalgebraic accounts deferred to appendices. The second (from Section 8 onward) asks how records are *formed* and *admitted*: the two-axis formation decision and its five operations, a learnable controller and a replay-calibrated utility estimator, the long-horizon social ladder (exchange harmful $\rightarrow$ temporal decay necessary-but-insufficient, with the receiver-relative form strongest among tested clocks $\rightarrow$ receiver-side admission), and a single integrated runtime evaluated under equal budgets, adversarial provenance, and continuous and VMAS substrates. The narrative is one arc: a shared archive fails, minimal admissibility mechanics help, the system learns what to retain, foreign testimony is validated on the receiver's own visit, and collective memory becomes net-positive at horizon.

2 Memory substrate and task

Each agent observes a grid environment and emits semantic place evidence such as `water_source`, `hazard_edge`, and `open`. A local memory stores place records, tag confidences, visit counts, freshness, and transition statistics. Planner queries retrieve candidate places by matching required, preferred, and penalty tags. A selected candidate is materialized into a concrete cell and executed by a local controller.

The multi-agent task uses a 12×10 grid. Each agent must reach a cell tagged `water_source`. Targets are scarce: $N\in\{3,5,8\}$ agents, $T\leq N$ targets, three layouts, three hazard densities, and 40 seeds for the main cadence sweep. Every run logs Q/R/M/C events per agent:

$$Q^\star \rightarrow R^\star \rightarrow M^\star \rightarrow C^\star,$$

where $M^\star$ means a semantic candidate was locked as an actionable target and $C^\star$ means the target was reached. These logs are not the architectural contribution; they let us see where each topology fails.

3 Four collective-memory topologies

The four topologies share the same local memory and planner. They differ only in where merge happens.

Independent. Each agent owns a private event store and concept graph. There is no communication path.

Shared bus. All observations are appended to a single global event bus. One shared concept graph is rebuilt from the union of all events.

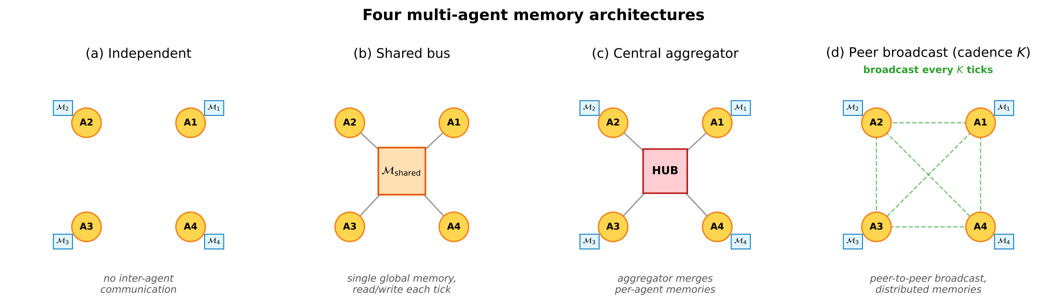

Figure 1: Four memory topologies. **(a)** Independent: per-agent private memory, no exchange. **(b)** Shared bus: one global memory, every agent reads and writes each tick. **(c)** Central aggregator: private memories plus a hub that merges snapshots into a consensus report. **(d)** Peer-to-peer broadcast: private memories with snapshot exchange every K ticks and per-agent merged views.

Central aggregator. Each agent owns a private concept graph; a central consensus layer pulls snapshots and computes one trust-weighted report for everyone.

Peer broadcast. Each agent owns a private concept graph and a private merged view. Every K ticks, agents broadcast snapshots; each agent merges its own memory with the latest snapshot from each peer.

The key distinction is not the clustering primitive. It is the level at which that primitive is instantiated:

$$\begin{array}{rcl} \text{independent} & : & N \text{ graphs, } 0 \text{ mergers} \\ \text{shared bus} & : & 1 \text{ graph, } 1 \text{ implicit global merger} \\ \text{central} & : & N \text{ graphs } + 1 \text{ central merger} \\ \text{peer} & : & N \text{ graphs } + N \text{ private mergers.} \end{array}$$

4 Cadence-dependent materialization–completion trade-off

This section restates, for self-containedness, the cadence result established in the companion Q/R/M/C study (Anonymous, 2026a); the sweep and its bottleneck-shift finding are that paper's contribution, and here they serve as the established starting point that motivates the CSM design. The peer topology has one explicit parameter: broadcast cadence K. The main sweep crosses four architectures and eight peer cadences $K \in \{1, 2, 4, 8, 16, 32, 48, 64\}$ for 35,640 runs. Figure 3 summarizes the result.

Fast peer broadcast and centralized sharing live in the M-saturation/C-collapse regime: agents can lock targets reliably, but many locks point to targets that peers also pursue. Independent and slow peer broadcast live in the M-deficit/C-saturation regime: fewer agents lock onto good targets, but those that do are less likely to collide with peer occupancy.

Across all peer runs, Spearman of $P(M^\star | R^\star)$ vs K is -0.608 and of $P(C^\star | M^\star)$ vs K is $+0.420$ (both $p < 10^{-4}$). The stronger signature is visible at fixed cadence: the within-cadence Pearson correlation between the M and C links peaks at -0.61 for $K{=}4$ and decays toward noise as K grows. Cadence therefore does not simply tune "amount of communication." It moves the bottleneck between semantic commitment and shared-resource completion.

5 Minimal collective semantic memory

The cadence sweep shows that fixed-rate snapshot exchange exposes only one control surface. A collective semantic memory should expose at least three:

Merge. How peer evidence about the same semantic place is combined.

Trust. How evidence from different peers is weighted when it conflicts or becomes unreliable.

Staleness. How old evidence decays before it influences retrieval.

The minimal CSM uses peer broadcast at $K{=}8$ and overlays three rules:

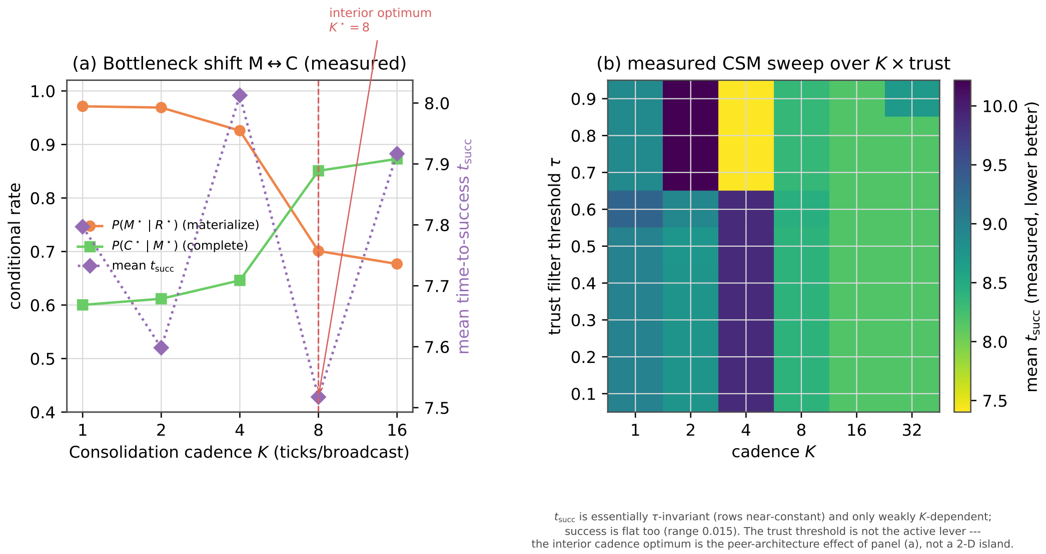

Figure 2: (a) The materialization–completion bottleneck shift and the interior optimum of conditional (success-only) time-to-success at $K^\star$=8, measured on the 35,640-run peer-broadcast sweep of the companion Q/R/M/C study (Anonymous, 2026a): $P(M^\star|R^\star)$ falls and $P(C^\star|M^\star)$ rises with K, while mean t_{succ} has an interior minimum. (b) The measured CSM sweep over cadence K and trust threshold τ: mean t_{succ} is essentially τ-invariant and only weakly K-dependent, and mission success is flat in (K, τ) (range 0.015). The trust threshold is *not* the active lever in this regime — the interior operating point is the cadence effect of panel (a), not a two-dimensional optimum island.

Merge. Per-place score equals own evidence at weight 1 plus peer evidence at weight $\text{trust}_i \cdot e^{-\alpha \cdot \text{age}}$, with $\alpha = 0.05$. Inclusion threshold $\tau = 0.30$.

Trust. Per-peer EMA on retrieval-success consistency. Peers whose latest snapshot supports the locally locked target gain trust; others decay toward 0.4. Update rate $\beta = 0.10$.

Staleness. Each broadcast snapshot is timestamped; merge weight decays as $e^{-\alpha \cdot \text{age}}$. Snapshots older than $10K$ ticks are pruned.

The implementation is a drop-in replacement for fixed-cadence peer memory: `experiments/collective_semantic_memory/csm_memory.py`.

5.1 Results

We compare CSM against fixed-cadence peer broadcast at $K \in \{1, 4, 8, 16, 64\}$ on scarcity cells $(N, T) \in \{(3, 2), (5, 3), (8, 5)\}$, three layouts, three hazard densities, and 20 seeds: 3,240 runs total.

Table 1: Minimal CSM vs fixed-cadence peer broadcast on the scarcity benchmark. CSM's lower 95% bootstrap CI bound exceeds every peer-K upper CI.

Architecture	$P(M^\star \vert R^\star)$	$P(C^\star \vert M^\star)$	Success rate (95% CI)
peer (K=1)	0.991	0.586	0.577 [0.572, 0.586]
peer (K=4)	0.955	0.623	0.581 [0.572, 0.590]
peer (K=8)	0.730	0.871	0.599 [0.591, 0.607]
peer (K=16)	0.697	0.892	0.597 [0.589, 0.606]
peer (K=64)	0.684	0.900	0.601 [0.594, 0.609]
CSM (minimal)	**0.782**	**0.825**	**0.617 [0.610, 0.624]**

CSM occupies a third regime. Fast broadcasts have excellent materialization but poor completion; slow broadcasts have the reverse. CSM keeps both conditional links above 0.78. The absolute success gain is modest, +1.7 to +4.0 percentage points, but the stage decomposition shows why it

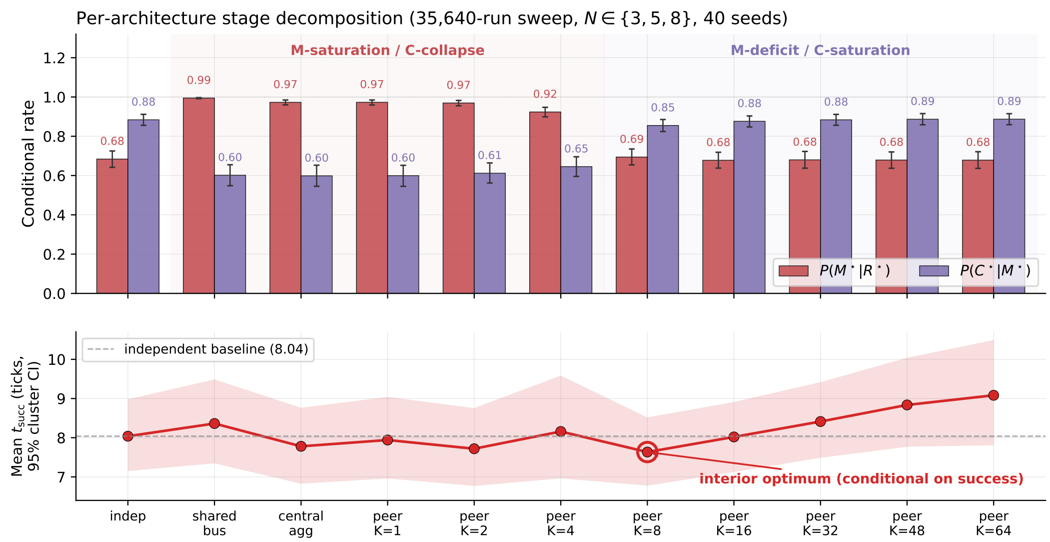

Figure 3: Conditional rates and mean time-to-success from the 35,640-run sweep. Fast sharing improves target materialization but harms completion through contention; slow sharing has the opposite profile. Among successful episodes, mean time-to-success has a shallow interior minimum at K=8; under a censoring-aware expected cost the advantage disappears (companion Q/R/M/C study).

is meaningful: the architecture changes the joint M×C operating point rather than merely moving along the fixed-cadence curve.

Mechanism: gated consumption rather than unrestricted sharing. Provenance annotation of the same runs gives the mechanism. CSM does not share *better* than fixed-cadence peers — it *refuses to act on stale or untrusted foreign evidence*. The inclusion rule of Section 5 implies a closed-form horizon beyond which a peer's evidence can no longer form an intention, $\mathrm{age}_{\max}(\mathrm{trust}_i) = \alpha^{-1}\ln(\mathrm{trust}_i \cdot c/\tau)$, and this horizon is empirically exact: measured intention-formation cutoffs match the formula tick-for-tick across trust levels (23/23.05, 18/18.59, 12/12.84 at $\mathrm{trust} = 1.0/0.8/0.6$), including a registered-prediction hold-out on six never-run cells ($6/6$ exact integer breakpoints) and a discrete trust-flip: $\mathrm{trust} \leq \tau/c$ zeroes foreign-evidence commitments entirely. Fixed-cadence peers have no such gate — their foreign evidence never expires — which is precisely why their completion collapses when sharing is fast. The full provenance-conditioned analysis appears in the companion route-transfer study (Anonymous, 2026b); here it serves as the mechanistic reading of Table 1.

6 Place identity from meaning: removing the shared frame

Everything above — and the categorical reading below — assumes a shared spatial basis X: 'the same place' means 'the same (x, y) in a frame the environment hands to every agent'. This section removes that assumption and reports what it was hiding. Each agent now holds a *private* frame (the true grid under a secret translation, and, in the final experiment, a secret 90°-multiple rotation); broadcast snapshots carry the sender's private coordinates, which are meaningless to the receiver until it decides what the sender's places *mean*.

Mechanism. A *place fingerprint* is the local semantic constellation around a visited cell: which content tags (hazards, walls, water) sit at which relative offsets — translation-invariant and coordinate-free. Two agents match places by *mutually unique* fingerprint correspondence (each side is the other's sole candidate; one-directional uniqueness aliases distant look-alikes into the receiver's neighborhood, and border-only patterns are excluded as descriptions of the world's shape rather than of a place). The inter-agent frame is the modal translation of ≥ 3 agreeing matched pairs; under rotations, one hypothesis per 90° reuses the same aligner and the winner must dominate the runner-up's consensus

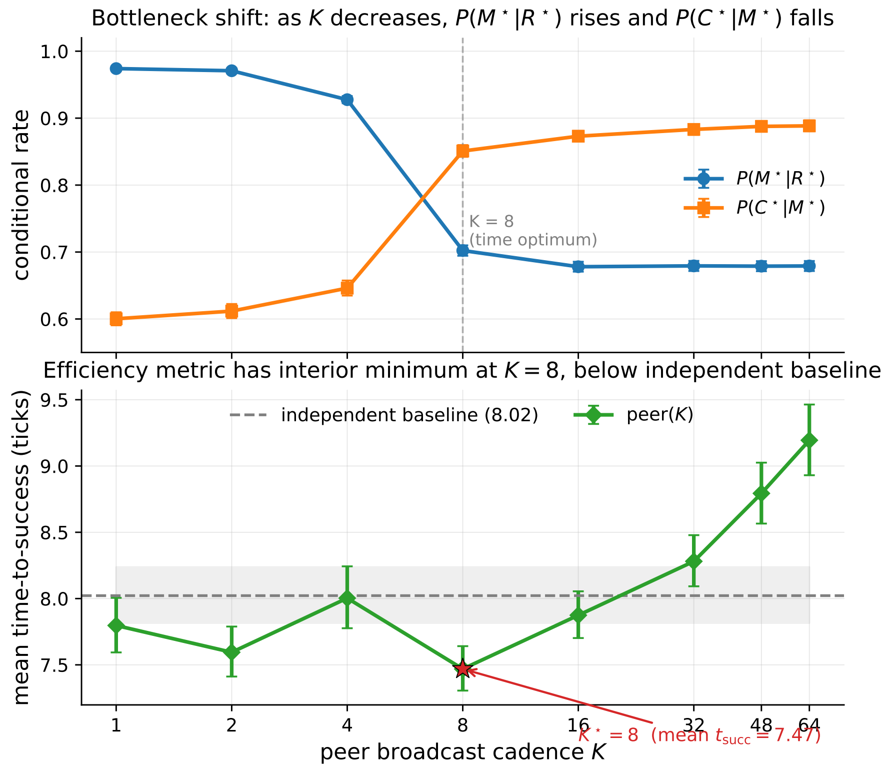

Figure 4: Bottleneck shift across peer cadences. $P(M^{\star}|R^{\star})$ falls and $P(C^{\star}|M^{\star})$ rises as K grows. The practical efficiency metric, mean time-to-success, is minimized at $K{=}8$.

by $2\times$. Foreign evidence — including places the receiver has never seen — is then transported through the recovered frame.

Results. Registered before execution; full protocol in the companion route-transfer study (Anonymous, 2026b).

1. **Exact recovery.** Witness–traveler: the secret offset is recovered exactly in 100% of landmark-world runs, and under translations and quarter-turn rotations (the discrete rigid-frame family $\mathbb{Z}^2 \rtimes C_4$) in $80/80$ rotation$\times$offset$\times$seed combinations, with 100% completion on foreign-committed locks.
2. **Failure asymmetry.** Where identification is impossible — a featureless region, or a $180°$-symmetric layout in which the frame is unrecoverable *in principle* — semantic identity makes *zero* commitments (fails closed), while coordinate identity keeps confidently locking wrong cells (100% wrong locks; fails open).
3. **The full stack.** With $N{=}4$ agents in distinct secret frames, alignment develops online (onset in 100% of episodes, mean tick ≈ 15) and recovers 88% of the shared-frame performance gap (success 0.444 vs. 0.456 oracle vs. 0.350 coordinate); under randomly sampled translations and quarter-turn rotations every semantic commitment points at real water $(8/8)$ while the coordinate stack produces 992 phantom locks (0 real).
4. **The gate is orthogonal.** Re-running the $\mathrm{age}_{\max}$ analysis above with the CSM inclusion rule ON TOP of semantic identity, under a frame offset and under a $90°$ rotation, reproduces the SAME integer breakpoints ($20/14/9$; $\mathrm{bp} = 20$ rotated) — the trust$\times$staleness gate and frame recovery are independent layers.

These results constructively address the previously stated shared-frame limitation: place identity can be carried by meaning alone, it composes with merge/trust/staleness unchanged, and — unlike the coordinate convention it replaces — it knows when it does not know.

The categorical indicator: adjunction round-trip defect. Frame recovery is exactly the operation the categorical reading of §A needs once the shared basis X is dropped: with private frames each agent carries its own place category, and communication is a pair of recovery functors $F = \textsc{align}(A, B)$ (the sender B's frame expressed in receiver A's coordinates) and $G = \textsc{align}(B, A)$. If the two are genuinely adjoint — the frames are glued — the round trip $\eta_X : X \to G(F(X))$ is the identity, so the *adjunction defect*

$$\mathrm{def} = \|F.\mathrm{offset}{+}G.\mathrm{offset}\|_1 \text{ (translation)}, \qquad r_F{+}r_G \equiv 0 \pmod 4 \text{ and } \|R_{r_G}F.\delta{+}G.\delta\|_1 \; (\mathbb{Z}^2{\rtimes}C_4),$$

is zero (Mac Lane, 1998). Measured on the recovery method above (`experiments/warp/alignment_defect.py`, 3 offsets $\times$ 10 seeds per cell): the defect is *exactly* 0 in $30/30$ runs for translation frames and $30/30$ for each of the four $90°$ rotations — the recovered transformations are mutual inverses within the implemented discrete transformation family, yielding zero round-trip defect. Where recovery must fail (an agent that has observed only a featureless interior band), the observations are insufficiently discriminative to identify a unique admissible alignment: several transformations are equally consistent, so the runtime leaves the alignment edge undefined in $30/30$ runs and refuses to perform the gluing, with *zero* spurious gluings. The exact discrete aligner therefore either recovers a mutual-inverse pair (def$=$0) or declines to instantiate the edge — and never produces a confident wrong gluing; "failing closed" above is precisely *declining a non-identifiable alignment*. This is the quantity the categorical model computes that a re-description does not. The round-trip defect is a *pairwise* witness: zero defect indicates that the recovered transformations are mutual inverses on the observed substructure. Global coherent gluing additionally requires cycle consistency across overlapping alignment paths ($F_{jk} \circ F_{ij} = F_{ik}$), formalized in §A; it is this gluing — not a shared coordinate convention — that the merge silently presupposed.

7 From distributed mechanics to memory formation

We introduced a minimal collective semantic memory in which merge, trust, and staleness are represented as separate, inspectable mechanisms. The experiments show that collective performance is not a monotone function of shared-information volume or communication frequency. The effective component is the admissibility gate: agents benefit when foreign evidence is merged with source information, weighted by trust, and removed from action authority as it becomes stale.

The same architecture can operate without a shared coordinate frame when local place records are aligned through semantic fingerprints. The formal analysis characterizes this alignment as a coherent gluing of local ordered memories and clarifies the distinction between a single population-spanning consensus component and several sub-consensuses. The coalgebraic interpretation complements this static description by representing the collective as an ongoing behavioral process.

The LLM experiments reinforce the central systems conclusion. Raw context can contain all historical evidence while still failing to resolve which information is current or which records refer to the same world entity. A consolidated graph supplies these relations explicitly and reduces the active context required by the model. The remaining problem is not how to coordinate a memory that already exists, but how agents should learn which observations, distinctions, and route structures deserve to become long-term memory. The second half of the paper addresses this formation problem.

8 Formation and admission: learning what to retain

Memory formation as a two-axis decision (counterfactual utility times simplicity of analogy): measured deterministically, learned by a contextual bandit that outperforms the hand-designed rule, evolved to a finite budget, and priced against seven policies over heterogeneous embodiments.

The preceding sections specify how structured memories are aligned, merged, and consumed; we now address how observations become durable memory records. This is a separate problem. An effective long-term memory must distinguish between repeated evidence, useful exceptions, novel schemas, short-lived observations, and records whose apparent similarity would produce an incorrect action if merged.

The need for an explicit formation mechanism is already visible in the Q/R/M/C diagnosis. In the single-agent benchmark, candidate ranking is accurate when candidates exist, but the memory frequently contains no eligible record. The limitation is therefore not only retrieval; it is the preceding process that determines which observations are consolidated into queryable structure. In a collective, the same formation decision additionally affects communication cost, cross-agent transfer, and the possibility of propagating stale or misleading evidence.

We formulate memory formation through two conceptually distinct decision dimensions (they are not claimed to be statistically independent). The first is counterfactual marginal utility: the expected change in task or team cost produced by adding a candidate record to the current memory. The second is structural assimilability: the cost of mapping the candidate onto an existing schema while preserving decision-relevant relations. Utility answers whether the record is worth retaining. Analogy answers whether it should update an existing schema or create a distinct one. A single importance, novelty, or similarity score cannot represent both questions.

The resulting controller selects among five operations. The decision table below is *formal*: its rows are the mutually exclusive, exhaustive branches of the implemented rule (utility threshold U_{thr}; signature-overlap threshold; decision-distortion threshold D_{thr}), so every incoming record falls into exactly one row:

Utility	Analogy	Conflict	Operation
high	none (no matching schema)	—	**NEW_SCHEMA**
high	complex (overlap below threshold)	—	**NEW_SCHEMA**
high	simple	consistent ($D<D_{\mathrm{thr}}$)	**MERGE**
high	simple	conflicting ($D\geq D_{\mathrm{thr}}$)	**EXCEPTION**
low	none	—	**DROP**
low	complex	—	**DROP**
low	simple	consistent	**REPEAT**
low	simple	conflicting	**REPEAT**

Two consequences of the implemented rule are worth making explicit. The conflict predicate is evaluated only when utility is high: a conflicting but useless candidate updates statistics (REPEAT) rather than earning an exception — EXCEPTION is reserved for evidence worth acting on. And conflict presupposes a simple analogy (distortion is measured against a matched schema), so the cell "complex $\wedge$ conflicting" is unreachable by construction.

The exception operation is essential. A useful observation that contradicts a stored schema should not overwrite the original evidence, but it should not be discarded either. It is retained as a context-dependent counterexample and resolved during later retrieval or schema splitting. This design is supported by an immutable episodic layer and by provenance links from schemas to their source episodes.

Necessity of the operations, by ablation. Each operation was removed in turn and the ablated memory run on paired two-world streams — an original world A and a corrupted variant V — on 48 worlds from seed blocks fully disjoint from every development seed (registered predictions written before the run; artifact `tmp/cluster/operation_ablations/`). Removing EXCEPTION in favour of last-write-wins merging destroys the original world (cost_A 54–58 vs. full's 12–13; success_A $1.0 \to 0.0$; corrupt-merge events in every world); removing it in favour of DROP destroys the corrupted world instead (cost_V 63–74 vs. 19–20): the exception cell is exactly what holds the two-world Pareto point. A binary merge-or-drop policy is catastrophic on transfer worlds B/C (100–119 vs. 12–16). Removing REPEAT leaves costs untouched while bloating memory and erasing support statistics; removing NEW_SCHEMA leaves cost intact but corrupts structure — schemas collapse into a single parent whose certificate accumulates false failure entries, the damage surfacing in trust bookkeeping rather than in path cost. Honest accounting of the registered verdicts: the population-level effects above are large and stable across all three seed blocks, but the *per-world universal* form of the exception prediction (strictly lower two-world max-regret in every single world) fails in 2 of 48 worlds, where last-write-wins happened to carry lower regret (8 vs. 10; 5 vs. 8) — and one seed block's streams contained too few novel events to exercise the NEW_SCHEMA verdict at all. We retain the universal predictions as registered, report them as $46/48$ and $2/3$ blocks, and state the necessity claims at the population level.

The empirical program proceeds in four stages. First, deterministic replay tests the five-way formation rule under controlled conflicts. Second, a contextual bandit and a terminal-reward learner evaluate whether the formation policy can be recovered from coordination feedback rather than fixed by the designer. Third, evolutionary search and structural matching test whether the route grammar and analogy mechanism can be compressed without losing decision quality. Fourth, an integrated multi-agent runtime evaluates receiver-side admission, world-time staleness, equal budgets, perceptual noise, adversarial provenance, continuous transfer, and LLM active-context reconstruction.

The formation-and-admission half contributes the following:

1. We define a two-axis memory-formation model that separates counterfactual usefulness from structural assimilability and includes an explicit exception operation.
2. We specify a conditional memory record with provenance, per-role utility, applicability conditions, analogical mappings, and failure cases, together with a receiver-side admission process.
3. We show that the formation policy is learnable and that its parameters can be optimized within a fixed grammar family.
4. We evaluate the method over horizons of up to 1000 episodes, identify staleness as the principal long-horizon failure of consolidation-only policies, and measure the contribution of freshness and trust-aware conflict handling.
5. We integrate formation, communication, quarantine, admission, alignment, route consumption, reservation, rupture, and rollback into a single runtime and compare it with independent, raw-history, heuristic, and structurally ablated memories under equal budgets.
6. We evaluate utility estimation without oracle replay, causal landmark promotion from a fixed candidate set, graph-based analogy, LLM context reconstruction, adversarial provenance, and fail-closed safety on discrete, continuous, and VMAS substrates.

Throughout, *receiver-side admission* means that a foreign certificate remains quarantined until its utility is evaluated under the receiver's current role and world version on the receiver's own visit; only records that pass this local check gain action authority.

The scope is that of an end-to-end research prototype. The utility estimator is calibrated against exact counterfactual replay in controlled environments; its validity in external stochastic systems remains open. Landmark promotion operates over a predefined candidate superset rather than generating new perceptual concepts. The grammar parameters evolve within a fixed representation family. These restrictions are retained explicitly in the claims and limitations.

9 The model

Counterfactual marginal utility. For memory item m, agent i, intent ι and current memory M:

$$U_{i,\iota}(m \mid M) \;=\; \mathrm{cost}_i(M, \iota) \;-\; \mathrm{cost}_i(M \cup \{m\}, \iota),$$

measured by *deterministic replay* with the item switched on and off — the source-masked replay device of the warp analysis, generalised from "foreign evidence on/off" to "this memory on/off." Utility is contextual (per agent and intent) and marginal (relative to what is already stored): a duplicate of a stored schema has $U = 0$ *by construction*, so repeats are detected without any similarity heuristic. The team-level version subtracts collision, communication and false-transfer costs. Exact replay is used as an oracle only during controlled calibration; at runtime the deployed controller uses the estimator $\widehat{U}$ (Section 17, UE1) rather than any simulator oracle.

Analogy cost. A new episode e maps onto a stored schema s through an analogical map ψ with cost

$$C_{\mathrm{analog}}(e, s; \psi) = L(\psi) + \lambda_d \, D_{\mathrm{decision}}(e, s) + \lambda_a \, \mathrm{Defect}(\psi), \qquad \mathrm{assim}(e, s) = e^{-C_{\mathrm{analog}}},$$

where L counts structure outside the correspondence (here: couplets outside the signature LCS), D_{decision} measures how differently the two songs *decide* — operationalized as the distance between their end-to-end beat displacements, i.e. where they ultimately send the walker — and Defect penalises broken relational structure. Descriptive similarity and decision agreement are different quantities, and D_{decision} captures the second; this is the rate-distortion intuition of Zou et al. (2026) imported into a multi-agent, structural setting.

The memory unit is a conditional program, not a fact. We fix a single canonical record for the whole part,

$$m = (G, C, E, U, P, T, F, A, H),$$

with G the relational graph/song, C applicability conditions, E expected effects, U the role-conditioned utility profile, P provenance (references to immutable source episodes), T observation and world time, F failures and exceptions, A admission and authority state, and H alignment requirements (what a valid correspondence would have to satisfy). Here A denotes admission and authority state; it must not be confused with the temporary analogical map ψ_m (below), which is reconstructed during consumption and never stored. Indeed, the concrete analogical map ψ_m and the inter-frame correspondence $\Gamma_{ij,t}$ are reconstructed by the receiver at consumption time and are not transmitted as authoritative state. On exchange, a schema travels with a **utility certificate** containing applicability conditions, observed ΔV, uncertainty, evidence identifiers, alignment requirements H, and recorded failures; the receiver transports the song through its own mapping and recomputes its own utility. For heterogeneous embodiments this is essential: a route through a hazard field can have $U_{\text{robust}} > 0$ and $U_{\text{fragile}} < 0$ simultaneously, so a memory item carries a utility function over roles, not one weight.

Four storage layers. (1) an *immutable episodic store* — source episodes are never rewritten (continuous rewriting first helps and then hurts (Zhang et al., 2026b); primary evidence anchors reconsolidation); (2) the *analogical schema graph* — songs, landmarks, exceptions, governed by the two-axis matrix; (3) a *procedural layer* for schemas promoted to policy fragments; (4) a *coordination layer* for social rules (reservations, roles, trust, transfer conditions). An agent's active context is a reconstruction — relevant schemas, minimal supporting episodes, current exceptions and the coordination contract — not a replay of history.

9.1 The runtime as one algorithm

Every experiment below is a configuration of one per-agent loop; we state it once so the method is reconstructable from the paper alone. A record is the canonical $m = (G, C, E, U, P, T, F, A, H)$ of §9 — song graph, applicability, expected effect, per-role utility, provenance, observation/world time, failures and exceptions, admission/authority state, and alignment requirements H. The concrete inter-frame correspondence $\Gamma_{ij,t}$ is re-derived at consumption and never stored.

Algorithm 1 (Songline Memory Runtime).
In: observation o, intent ι, local memory M, quarantine Q.
1. extract candidate song from o; 2. estimate marginal utility $\widehat{U}$ (§17, UE1); 3. find structurally closest schema (analogy §15); 4. compute analogy cost + decision distortion; 5. choose DROP / REPEAT / MERGE / EXCEPTION / NEW_SCHEMA; 6. append to immutable episodic store; 7. update schema graph; 8. build provenance + utility certificate; 9. broadcast on cadence, within wire budget; 10. place received records in Q; 11. on visit, validate world-time, role, local utility; 12. promote Q-record to retrieval/planning/action authority; 13. retrieve an admissible song (support$\times$staleness order); 14. recover landmark correspondence (consensus $\geq k$ + loop closure); 15. reserve target/route; 16. execute safe prefix (≥ 5 couplets verified); 17. detect rupture; 18. commit / replan / rollback.

Quantities in one place. Marginal utility $U_{i,\iota}(m|M) = \mathrm{cost}_i(M, \iota) - \mathrm{cost}_i(M \cup \{m\}, \iota)$; analogy cost $C = L(\phi) + \lambda_d D_{\text{decision}} + \lambda_a \mathrm{Defect}(\phi)$; retrieval score support $\cdot\, e^{-\alpha\,\mathrm{age}}$ (trust-flip $\times 0.25$ on an excepted parent); world-clock admissibility $\mathrm{ver}(m) \geq \text{known\_ver(fam)}$; rupture index $=$ $\arg\min_e \mathrm{trust} \cdot e^{-\alpha\,\mathrm{age}(e)} < \tau$; objective $w_s\,\mathrm{succ} + w_t(-t) - w_b\,\mathrm{bits} - w_c\,\mathrm{collisions}$ (weights w_s, w_t, w_b, w_c distinct from the staleness rate α above and from the conditional product Φ of the companion Q/R/M/C evaluation study (Anonymous, 2026a)).

Hyperparameters (frozen; test seeds never used to tune). utility threshold U_{thr}=5 (validation, U1); analogy share ≥ 0.4 and distortion ≥ 3 (validation, U1); staleness α=0.05, τ=0.30 (fixed; the trust–staleness constants of the companion route-transfer study (Anonymous, 2026b)); song budget $v_{\max}$=10 (*evolved*, U3); anchor consensus k=3 and safe-prefix depth 5 (safety design, N1/C1). Selection method and per-parameter sensitivity are tabulated in Appendix D.

10 U1: the matrix, measured deterministically

Eight valid grid-world families, five candidate memories per family — one per matrix cell, in sequence: a useful novel song (NEW), its exact duplicate (REPEAT), a song from a structurally different world (NOVEL), a song to a goal irrelevant to the water intent (IRRELEVANT), and a song from a variant world whose water has been secretly moved (CONFLICT). Two single-axis baselines run on the same stream: similarity-only (store if novel, last-write-wins merge if similar) and utility-only (store if useful, consume newest-first). Four predictions were registered; the first registration failed on two construction defects (an "irrelevant" goal that accidentally lay en route to the water; consumers spawning inside walls), both documented, and the corrected registration passed $4/4$:

- **Repeats are free by construction:** the duplicate's marginal utility is exactly 0 in every world; final memory is 3 schemas vs. 5 for append-everything.
- **Similarity is not a consolidation criterion:** the similarity-only baseline incorporates the irrelevant song and silently corrupts the original world by overwriting its schema with the conflict variant (62.6 steps and phantom-first, vs. 12.3 **steps first-try** for the two-axis memory).
- **Exceptions beat both overwrite and recency:** with the exception stored, the variant world costs 20.5 vs. 70.4 without it, while the original stays first-try correct; utility-only's recency order phantoms first on the original (24.0 steps).
- **Certificates are recomputable:** a receiver at a different position recovers positive utility from the certificate's evidence in every world.

11 U2: the matrix is learnable — and improves itself

A LinUCB contextual bandit replaces the hand thresholds. Features: $(U, \text{share}, D, \text{novelty}, \text{memory size})$; actions: the five operations; reward: the *measured* cost change of the memory on a two-world probe (the candidate's world and the nearest schema's origin world — the second term is what teaches the controller not to overwrite), minus a small storage tax.

The first registration failed on two evaluation defects that are themselves findings. The final battery was circular (each policy was scored on the worlds it chose to store, making random indistinguishable from designed), and two pairs of actions were *reward-equivalent*: NEW and EXCEPTION had the identical structural consequence, as did DROP and REPEAT — and the bandit recovered exactly that quotient, confusing actions strictly *within* equivalence classes while learning the drop cell at $40/43$. Bookkeeping without consequences is unlearnable. Version 2 made the battery independent and gave support a consequence (consumption order). Result:

	learned	hand matrix	random
held-out battery cost	**78.1**	83.5	105.9

The learned policy reduces task cost by 6.5% relative to the hand-designed policy. Its majority actions agree with the hand matrix in $3/5$ cells (a registered miss); the disagreements are two context-dependent equivalences (merging a duplicate $\equiv$ repeat; repeating junk $\equiv$ drop) and one *principled, profitable deviation*: where the hand matrix says MERGE, the controller appends a new version and lets support ordering arbitrate — it independently discovered the immutable-store principle (§9), the same lesson the consolidation literature reaches from failure analyses (Yang et al., 2026; Zhang et al., 2026b).

12 U3: optimization of grammar parameters within a fixed family

An evolution strategy (30 generations $\times$ 16 genomes) selects the grammar $\theta = (\text{couplet gap}, \text{share threshold}, \text{distortion threshold}, \text{utility threshold}, \text{couplet budget } v_{\max})$, with fitness measured on maps the genome never sang on (cost + λ·bits). Both registered predictions passed: the evolved genome matches the hand grammar on hold-out within registration tolerance, and it selects a *finite* budget ($v_{\max} = 10$) with no cost regression — bounded rationality obtained by selection, not imposition. Beyond registration: selection converged to the neighborhood of the hand-designed thresholds (share $0.74 \to 0.46$ vs. hand 0.4; distortion 3.9 vs. hand 3) while choosing denser couplets and a lower utility bar — "store more, but shorter." The designed constants of the matrix sit near a selected parameter region rather than being arbitrary.

13 U7: seven policies, two bodies, three horizons

The decisive comparison runs one mixed episode stream (new families, exact repeats, appearance variants with resampled hazard texture over persistent walls-and-water structure, and water-moved conflicts) through seven team-memory policies, consumed by two embodiments — *fragile* (hazards cost 12 steps) and *robust* (hazards cost 1, steps cost 2) — and evaluates on the families' latest true worlds plus fresh appearance variants never streamed. Policies: raw long context (append all, newest-first), embedding-style signature similarity, recency-importance-relevance (Park et al., 2023), full-snapshot exchange, utility-only, analogy-only, and the full two-axis memory with role-profiled certificates.

Table 2: Mean fragile-role battery cost by stream length (12 seeds) and stored bits at 1000 episodes. All four registered predictions failed; the corresponding failure mechanisms are analyzed in the text.

Policy	10 eps	100 eps	1000 eps	bits @1000
raw long context	128.9	**53.3**	**139.8**	339,453
utility-only	130.5	53.6	144.6	162,107
two-axis (UCSM)	132.8	60.8	145.2	**153,774**
signature similarity	161.1	214.7	232.1	186,843
recency-importance-relevance	190.7	225.4	250.1	339,453
full snapshots	209.4	246.6	264.5	5,198,886

Failure analysis. The registered dominance, sublinear-bits, role-preservation and false-merge predictions all failed. Three mechanisms carry the content. *First*, the fail-closed identity layer makes bloat cheap in steps: stale or foreign songs simply produce no anchors and hence no targets, so the raw archive is filtered without an explicit storage penalty at consumption time and pays only in bits and matching compute. *Second*, sublinear memory growth is impossible without cross-family abstraction: in an open stream where 30% of episodes found new structures, consolidation caps within-family growth ($2.2\times$ fewer bits than raw, $34\times$ fewer than snapshots) but cannot compress novelty it has no schema for. *Third*, the long-horizon failure mode — phantom-first rates near 0.95 for every song policy at 1000 episodes — is a *staleness* cascade from accumulated conflicts, which no storage policy fixes by ordering alone; it is the native territory of the series' trust$\times$staleness law, which none of the seven policies implemented.

Results stable across horizons. At every scale, structural memory (identity + beats, whatever the consolidation rule) beats the mainstream heuristics — embedding similarity, recency-importance-relevance scoring (Park et al., 2023; Xu et al., 2025), and snapshots — by 1.7–$4.5\times$; and the two-axis memory is the cheapest in bits among song policies everywhere.

U7b: testing the staleness diagnosis. An eighth policy — the same two-axis consolidation consumed newest-first — was registered and run. At 100 episodes it *passes*: it matches or beats raw on both roles ($52.0/67.9$ vs. $53.6/68.9$) at 54% of raw's bits, closing the entire consolidation-overhead gap by ordering alone. At 1000 episodes it fails by $\sim 3\%$: under strong appearance drift, raw's redundancy retains residual value — older variants of a song still anchor where the newest one fails to. The program formula that survives: **consolidation buys bits, freshness buys success, and they compose at moderate horizons; the next lever is a real age gate and cross-family schemas, not better ordering.**

14 Addressing staleness, landmark selection, and abstraction

A second registered wave attacked the three most concrete gaps the first wave localized.

Staleness in consumption (U7c). The series' trust$\times$staleness contract was wired into consumption: schemas ordered by support $\cdot\, e^{-\alpha\,\text{age}}$, plus the discrete *trust-flip* — a schema that has accumulated an exception is demoted, so the counterexample is tried before its discredited parent. At 1000 episodes this closes the robust role entirely (214.2 vs. raw's 214.3) and cuts the fragile-role gap threefold (-0.9% vs. -3% for freshness ordering alone); the strict both-roles registration still fails — the failure is retained and analyzed: residual redundancy value under appearance drift survives even the gate. The composition ladder is now measured: consolidation alone -14%, $+$freshness -3%, $+$trust-flip -0.9%.

Causally promoted landmarks (E1). The causal loop *propose* $\to$ *ablate* $\to$ *measure* $\to$ *promote* was implemented and evaluated: starting from a feature superset including a derived candidate and deliberately harmful frame-variant junk (cell parity), the loop caught the junk causally ($\Delta = -0.67$; its harm manifested as refusals — zero fail-open across all ablation rounds) and reached a minimal vocabulary at perfect score ($0.33 \to 1.0$). The registered expectation that content classes would survive failed in the most instructive way: the surviving pair was $\{\texttt{wall}, \texttt{void}\}$ — the border feature that carries *zero* consensus mass in the attribution measurement turns out to be causally necessary as an *anti-aliasing guard* once the vocabulary is impoverished. Causal ablation and attribution mass are different quantities; the value of a feature is non-additive and contingent on the rest of the vocabulary.

First-order cross-family abstraction (X1). A shared signature codebook — constellations recurring across families stored once and referenced by index, lossless by construction — was added to the storage codec. Registered and passed: the codebook's bits growth ($7.91\times$ per decade) is smaller than the plain codec's ($8.42\times$), and the absolute saving grows with scale (-10% at 100 episodes, -16% at 1000): cross-family motifs exist and reduce storage through lossless reuse of recurring constellations. Sublinearity is approached, not achieved — full schema-level abstraction (slots and roles, not just shared constellations) remains the open half.

The LLM end-to-end cycle (L1). The full promised loop — observation $\to$ candidate memory $\to$ controller $\to$ schema graph $\to$ reconstructed active context $\to$ LLM action — was implemented on live models (the W4 division of labour: the LLM commits the semantic target, a deterministic waypoint primitive walks). Formation is deterministic UCSM over 4- or 12-episode histories in which the water is moved once or twice, so the stale targets hold up to a 10:2 majority of the raw transcript; a fresh LLM session then commits under three context arms (no memory / full transcript / schema graph with certificates and exceptions). Registered and measured: memory is needed (the no-memory arm is 0.0 everywhere); reconstruction beats replay — `llama3.1-8B` degrades on the raw transcript from 0.42 to 0.08 as history grows (conflict chains $0.17 \to 0.00$) while the active context holds 1.0 throughout at 6–12% of the transcript payload; `Qwen2.5-3B` reads the short transcript perfectly (1.0) yet drops to 0.375 on 12-episode chains, with the active context again at 1.0. A 4B thinking model fails the supersede chain in every arm — a capability floor below the task, reported for completeness. The transcript's failure mode is specifically the stale majority, not length (benign 12-episode logs stay readable to the stronger model): raw context reproduces the U7 phantom cascade in prompt space, and the reconstructed schema graph is its prompt-space staleness gate.

15 Extensions: structural matching, social exchange, and delayed credit

The remaining three gaps were evaluated at scale. One was resolved within the registered criterion, one produced a near-miss, and one returned the strongest negative result of the program.

Graph-matching analogy (G1). The LCS proxy was replaced by structural alignment: Needleman–Wunsch over couplets with partial node similarity (Jaccard over signature keys) and a *relational defect* — beat-chain consistency of the aligned pairs (do the two songs walk the same skeleton?). On appearance-heavy streams (300 episodes, 12 seeds) the graph controller recognises "the same structure in a new texture" that LCS blindly filed as new: 24% fewer schemas and bits (122 vs. 162 items; 42k vs. 55k bits) at cost within 5% on both roles and no safety price (phantom $+0.012$). The registered schema threshold ($\leq 0.75\times$) was missed by 0.01 — reported as a failed registration whose every substantive clause held.

Terminal credit (M1). The immediate-reward bandit was retired: an evolution-strategies meta-learner optimizes the same policy class against a *terminal-only* reward — one number per full stream, after dozens of interleaved decisions. On held-out streams the meta-learned policy beats both the densely-shaped bandit (-97.95 vs. -99.69) and the hand matrix (-99.95), and on hand-EXCEPTION candidates it takes store-class actions in 91% of cases with no replace-majority — the third independent recovery of the immutable-store principle (after the hand-designed controller and the bandit), this time from a single delayed scalar.

Long-horizon social exchange without admission (S1). Six agents of mixed embodiments, each with its OWN memory, acting simultaneously in a world whose families evolve globally, exchanging certificates on a cadence. At 20 episodes communication helps; at 300 episodes **independence wins outright** (fragile 148 vs. 170; robust 171 vs. 238 for every communicating arm). Mechanism: receivers accumulate hundreds of foreign schemas (741 vs. 221 items) whose world-referents drift stale faster than role-gating or recency can filter — broadcast time is not world time. Three of four registered predictions failed on this; the one that passed is reservations (S1.3: duplicate commitments to zero at no cost). This is the central empirical pattern of this paper — *the gate, not the exchange, carries the value of collective memory* — re-derived in the formation substrate: certificates without a world-clock staleness contract are not merely insufficient, they are net harmful at horizon, exactly as raw exchange was for the LLM collective in the companion route-transfer study (Anonymous, 2026b).

The world-clock contract (S2) — necessary, not sufficient. S1's diagnosis was implemented: every family carries a state version bumped when its world actually changes; certificates carry (family, version); version news travels in certificate metadata even when the schema is rejected; admissibility is gated on world clocks at both consumption and consolidation. On paired assignment streams (12 seeds, 300 episodes) the gate improves every communicating arm (robust 238 $\rightarrow$ 220, fragile 170 $\rightarrow$ 163) and reservations still eliminate duplicate commitments without additional task cost — but the S1 reversal does not reverse: independent agents beat the gated arm on *every* seed, and memories barely shrink (688 vs. 708 items). The gate recovers only $\sim$27% of the communication penalty. Two registered negatives now bracket the problem from both sides: message-time filtering (S1) and world-time filtering (S2) both fail to make social exchange net-positive at horizon. The residual mechanism is not temporal at all: it is *cross-family aliasing from sheer foreign volume* — hundreds of schemas about families the receiver rarely visits still partially match its local constellations and emit phantom targets. What the data now names is ADMISSION control, not staleness control: accept foreign knowledge only where it has demonstrated utility on one's own visits — the certificate is testimony, and testimony, the two negatives jointly say, is not admissible evidence.

Is a world clock actually necessary? Clock alternatives, measured. "Necessary" deserved a direct test against the standard alternatives, and the answer revises our own earlier phrasing (12 seeds, 300 episodes, paired streams; artifact `tmp/cluster/song_grammar/ca/`). Ordering-only logical clocks do *not* substitute: Lamport (194.9/258.8 fragile/robust) and vector clocks (195.0/258.9) are *worse than no clock at all* (169.8/238.1) — they order testimony but never age it out, and the extra admissible volume amplifies the aliasing failure. Family version numbers alone (162.5/220.4) approximate the world clock (159.2/211.8). The strongest temporal gate is *age since the receiver's own last corroboration* (129.1/154.5) — a purely local signal requiring no cross-agent time at all, which beats the world clock and even the independent baseline (148.2/170.8) in this setting, and which is precisely a weak form of receiver-side admission. The correct statement of S2 is therefore not "a world clock is necessary" but: *temporal decay of foreign testimony relative to the receiver's own evidence is necessary; a world clock is one adequate carrier of it, ordering-only clocks are not, and the receiver-relative form both outperforms global time and anticipates S3*. All arms here still lose to full admission (S3: 118/134).

Admission control (S3) — the reversal reverses. The principle the two negatives pointed at was implemented: every foreign certificate is QUARANTINED on arrival (version metadata still travels); when the agent visits family f, quarantined certificates about f are validated ON THE SPOT — the foreign song's marginal utility is replayed on the actual current world under the agent's own role; passers are admitted with their utility *measured, not told*, and are usable in that same visit; failers are discarded; knowledge about never-visited families never enters consumption. All four registered predictions pass (12 seeds, paired streams). **Social exchange is net-positive at horizon for the first time**: the full contract beats independence on every seed and both roles (fragile 118 vs. 148, -20%; robust 134 vs. 171, -21%). The mechanism is monotone — testimony $162/220 \rightarrow$ relevance-only admission $136/169 \rightarrow$ demonstrated-utility admission $118/134$ — hygiene finally materialises (admitted memory 233 vs. 688 items, $0.34\times$, with 592 junk certificates held harmlessly in quarantine), and reservations compose (zero duplicate commitments at $+0.6\%$ cost). The social ladder of the program is thereby measured end to end: exchange without contracts is harmful (S1), world-clock staleness is necessary but insufficient (S2), and admission by demonstrated utility on one's own visits — testimony is not evidence — resolves the evaluated social-memory setting (S3).

The component ladder, measured. The S1→S3 narrative is closed by a registered staged ablation on paired assignment streams (12 seeds, 300 episodes; artifact `tmp/cluster/song_grammar/cl/`): each rung adds one component, and every headline claim of this section re-derives as a rung difference. Mean cumulative cost (fragile/robust):

Rung	Configuration	fragile	robust
1	independent	135.5	143.6
2	+ raw exchange	151.3	192.2
3	+ world-time staleness	142.2	172.6
4	+ provenance/trust	141.7	172.9
5	+ quarantine (hold, no validation)	133.7	135.8
6	+ receiver-side admission	118.7	124.3
7	+ reservations (full)	118.5	123.8

Exchange hurts (rung 2), staleness gating recovers part (rung 3), quarantine-without-validation collapses to ≈independence (rung 5, as the theory of admission predicts: held-forever testimony is inert), admission produces the reversal (rung 6), and reservations add duplicate hygiene at no cost (rung 7: duplicate rate $0.015 \rightarrow 0.000$). Leave-one-component-out from the full runtime confirms the attribution and yields one honest negative: removing admission or quarantine breaks the system back to rung-3/4 costs ($141.4/171.7$ and $139.6/167.8$), removing staleness costs moderately ($119.6/132.1$), removing reservations returns duplicates without a cost change — and removing *provenance tracking* changes nothing measurable ($118.4/123.8$): in this substrate the trust weights are carried by admission itself, and provenance's value (auditability, adversarial attribution) is not visible in the cost metric. We report the null rather than claim otherwise.

The authority ladder, and why quarantine is not a deadlock. A foreign record here passes through four authority states shared with the companion route-transfer study (Anonymous, 2026b): *observed* (the sender's local evidence), *received* (quarantined: present, version metadata circulating, but inert), *provisionally executable* (permitted to guide bounded, reversible transport under an explicit validity-and-coordination contract), and *admitted* (durable authority after receiver-side validation). Quarantine restricts the *fourth* state, not the third. In this paper's long-horizon setting, agents revisit families in the course of their roles, so quarantined testimony is validated on visits that occur anyway and strict quarantine suffices; the apparent circularity — validation requires a visit, a visit seems to require acting on the record — does not arise. Where visits do not recur naturally, the companion study supplies the missing rung: bounded provisional transport (staleness gate, reservation, rollback, rupture boundary) carries the receiver into validation range without granting durable authority. The two papers describe two rungs of one ladder, not two competing principles.

16 One method: the integrated runtime and its benchmark

Every mechanism above was validated in its own controlled experiment; the central integration question is whether these mechanisms remain effective when composed. We therefore built **Songline Memory Runtime v1**: the one canonical record type $m = (G, C, E, U, P, T, F, A, H)$ of §9 (song graph, applicability, expected effect, per-role utility, provenance, time and world version, failures/exceptions, admission state, and alignment requirements; frame mappings $\Gamma_{ij,t}$ re-derived at consumption, never stored), one cycle from observation to execution, and every mechanism as a configuration flag — so the benchmark's arms and ablations are configurations of one artifact, not adjacent codebases. Wire and storage accounting uses the full codec (certificate 64, provenance 48, time/version 32, reservation 24 bits).

Single-run integration (R1). The ten-point completion test — route discovered in a private frame; song received; certificate held in quarantine; admission by the receiver's own measured utility;

landmark correspondence; route execution; rupture fallback after the world changes; reservation deferral; superseded versions never used — passes within a single execution of the integrated runtime, $10/10$ scenarios.

The integration benchmark (I1). Six agents of mixed embodiments, two intents (resource and rest), private frames, globally drifting families, 300 episodes, 12 paired test seeds (numbered $100+$, never used to tune anything; all thresholds frozen from the earlier waves). Group cost (fragile/robust): full Songline $\mathbf{120.0/125.1}$ dominates independent $(137.3/146.5)$, raw shared history $(160.9/204.4)$, graph-without-provenance $(146.3/174.5)$ and the song ladder without admission (146–157/174–196); it is simultaneously the only arm with zero duplicate commitments, the lowest phantom-first rate and a wire cost below raw's (55 vs. 69 KB). The advantage grows with scale — -11%, -13%, -14% at $N = 4, 6, 8$ (registered H3: **pass**) — and survives 10% and even 20% perception noise (-13–15%; the advantage clause of H2: **pass**).

Ablation matrix and failure modes. Removing landmarks or beats is catastrophic (success 0, cost worst: fail-closed by construction); removing admission costs $+22\%/+38\%$ — the component with the largest measured contribution, confirming S3; removing the world clock costs $+9\%$ on the robust role; removing reservations restores duplicate commitments; removing the utility gate bloats wire and memory by $\sim40\%$ at equal cost. But three ablations — provenance, exceptions, the immutable layer — are cost-free *in this substrate*: registered H1 therefore **fails** as stated, and the mechanism is worth the failure. Their value was demonstrated in the settings that exercise them (corruption on return in U1, gossip explosion in S1, audit and reconsolidation); the live-visit drifting benchmark never punishes their removal. In one artifact they are free insurance, not dead weight — but the composition claim must be scoped to the mechanisms the substrate actually stresses.

Utility without the oracle (UE1). A ridge estimator over runtime-available features only (anchoring, local plan distance, song shape, memory counts) reaches Spearman 0.989 and sign accuracy 0.97 against the true counterfactual on held-out worlds, and the full controller driven by $\widehat{U}$ alone matches the oracle version (cost ratio $0.965/0.977$) — the critical experiment: the system survives estimation.

Beyond rank correlation (UE1, extended). Because a high Spearman alone does not certify decision quality, the estimator is also evaluated on error, calibration, and regret (same frozen λ and thresholds; test seeds $100+$ untouched during any tuning; artifact `tmp/song_grammar/ue1_eval/`). On the held-out split: MAE 12.3 and RMSE 15.9 against $\mathrm{std}(U^\star)=89$ — so absolute values carry $\approx14\%$ error even where ranking is near-perfect; sign accuracy 0.98 (precision 0.978, recall 1.0 for $U^\star>0$); top-$5/10/20$ recall $0.82/0.88/0.90$; calibration monotone with weighted error 8.7, the main defect an over-estimate of weak records near the retention threshold. The decision-level metric is *policy regret*: retaining the top-B records by $\widehat{U}$ instead of by exact replay loses $\leq 0.3\%$ of the achievable value at every tested budget. Out-of-distribution slices: layout-variant and horizon shifts preserve all metrics (regret $\leq 0.7\%$); the honest weakness is the *role* shift (calibrated on the robust role, evaluated on fragile-role utilities): magnitudes are systematically under-predicted (MAE 33.8, calibration error 30), ranking survives (ρ=0.97, regret $\leq 2.3\%$) but top-5 recall drops to 0.60 — per-role calibration is required where absolute utility thresholds matter, and we record this as a limitation rather than average it away. All headline metrics replicate on a widened split (30 pools: MAE 10.7, ρ=0.991, regret 0.2%) and on a completely fresh seed block (200–230, never touched by any experiment in the program: MAE 10.7, ρ=0.990, sign accuracy 0.976, regret 0.3%).

The margin lesson (N1). Under independent false-negative $\times$ false-positive perception noise, coverage degrades gracefully ($0.96 \rightarrow 0.92$ at $10\%/10\%$: registered **pass**), but the fail-closed contract does NOT survive naive margin softening: with a plain cosine margin the fail-open rate reaches 0.21 inside the 20% envelope (registered N1.1: **fail**). Exact-equality matching was necessary in this configuration; the named fix is consensus-of-anchors verification (the identity layer's $\geq$ 3-pair consensus rule, which single-anchor song consumption bypasses) — margin machinery, not a lower threshold.

17 Integrated evaluation under equal budgets, adversarial provenance, and continuous transfer

We evaluate whether each mechanism produces its predicted failure mode in at least one targeted or integrated setting: at equal budgets the full method should beat the best baseline on unseen dynamic worlds without hurting safety, the advantage should survive noisy perception, private frames, heterogeneity and long-horizon drift, and removing a mechanism should reproduce its characteristic failure somewhere in the suite. We do *not* require every component to affect aggregate cost in every substrate — a mechanism whose value is shown in a specialized experiment (corruption for provenance, gossip for the immutable layer) need not also move the integration benchmark. We report each clause below.

Full resource accounting. Equal memory and communication budgets do not by themselves equalize *compute*, so every arm of the benchmark was re-run under full metering (30 test seeds, e=300, paired with the utility benchmark; artifact `tmp/cluster/song_grammar/resource_accounting/`): stored bytes, transmitted bytes (protocol metadata included), memory-update time, query latency, process CPU-seconds, and amortized formation cost per use. The utility-winning arm is not buying its advantage with hidden compute: the full runtime sits on the joint (cost $\times$ all six resources) Pareto frontier and even has *lower* query latency than independent memory (79 vs. 97 ms mean), because consolidation keeps the queried store small. Its real premiums are storage (624 KB/agent vs. 181 independent) and wire traffic (54 KB), reported as such. The compute-expensive arms are the unconsolidated retrieval-scan baselines (query 345–653 ms, CPU 1.6–2.9$\times$ the full runtime), while the top-3-capped graph baseline is the CPU-cheapest arm of all (30 CPU-s) at much worse task cost — structure is not intrinsically expensive; scanning is. One metering gap is disclosed: the learned-formation arm's counterfactual probes made its shards exceed the walltime (8 of 10 seeds per shard completed; its update cost, $\sim$870 s per seed, is itself the finding), and reservation traffic in this substrate is estimated, not carried on the wire.

Statistical footing. On the integration benchmark the full method beats independent on $12/12$ **paired test seeds** (mean team gain 13.6%, sign-test $p = 0.0002$), and its hazard contacts are *lower* (5.06 vs. 5.66 per visit) — the advantage does not buy speed with safety.

Equal budgets (B1). Capping wire and memory at the full method's own footprint (55/45 KB) and giving the raw-history baseline the same, the full method obtains $119.8/124.5$ versus $158.7/199.9$; and raw does not catch up at $2\times$ or even $4\times$ the budget ($159.3/201.1$ throughout). Raw's deficit is not a storage shortage that a bigger cap fixes — it is the absence of consolidation and admission. (On the continuous substrate raw is not merely worse but computationally intractable at the full horizon — $\mathrm{O}(E^2)$ matching over an unconsolidated store — a cost the wire/byte accounting alone does not capture.)

Adversarial provenance (P1). With one poisoning peer that corrupts its own songs and launders mutated foreign records under their original origin, the full method degrades $+1.2\%$ while the defense ordering holds (full $\leq$ no-provenance $\leq$ no-admission). The instructive negative: the unprotected arm barely degrades under poison *because it is already saturated with phantoms from ordinary drift* (159.7 clean). The dominant adversary of collective memory is world entropy, not a malicious agent — and admission, which gates on demonstrated utility at the receiver's own visit, absorbs the liar and the stale alike; origin-bound receipt blocks the laundering at no measurable cost.

Safety under noise (N1, revised). The fail-open bar $< 1\%$ is not met by a softened cosine margin (up to 0.21). The named fix — require a consensus of ≥ 2 mutually-matched anchors whose pairwise displacements agree with the beat chain (the identity layer's loop-closure rule, which single-anchor song consumption bypassed) — drives **fail-open to** 0.000 **across the entire** $20\%\times20\%$ **envelope**, trading coverage ($0.88 \rightarrow 0.58$ at $10\%/10\%$). Safety is bought with refusals, as the fail-closed contract demands.

A second substrate (C1). Formation and consumption were moved to a continuous box with real-valued feature positions, per-observation jitter, soft constellation matching (a positional tolerance

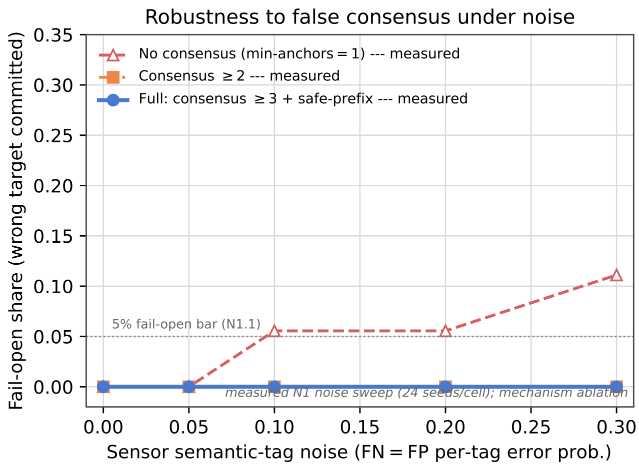

Figure 5: Fail-closed safety under sensor noise (measured N1 sweep, 24 seeds/cell, diagonal FN=FP). Removing the anchor-consensus mechanism (min-anchors $= 1$) lets the fail-open rate climb above the 5% bar as tag noise grows; anchor consensus ≥ 2 and the full consensus ≥ 3 + safe-prefix arm hold fail-open at 0.000 across the whole sweep — degrading into refusals (a coverage cost) rather than wrong commitments.

replacing exact key equality), unimodal anchor clustering with centroid refinement, and tolerance-based loop closure — the runtime otherwise unchanged. The advantage transfers on both roles and against both baselines — at a matched feasible horizon full $245.5/71.4 <$ raw $256.9/101.2 <$ (fragile) and $<$ independent $255.3/75.9$ (C1.3), and at the full horizon full $244.7/74.7 <$ independent $256.8/79.6$ (C1.1) — confirming the effect is not a grid artifact. Continuous social-scale safety needs a *three-layer* defense, each layer catching what the previous misses: anchor consensus (≥ 3 mutually-matched, loop-closed anchors) reduces per-transport aliasing ($0.79 \rightarrow 0.15$); single-commitment gating (walk only the most-anchored target, never a phantom candidate list) reaches 0.04; and a *safe-prefix verification* — the traveler physically walks the song's first five couplets and requires each observed constellation to match, so a foreign-family phantom that diverges from the actual terrain is refused before commitment — drives fail-open to $\mathbf{0.0014}$ on held-out test seeds ($554\times$ below the raw continuous rate, $10/12$ seeds exactly zero), *under* the 1% safety bar, while first-lock success *rises* (phantoms that used to be committed are now filtered). The prefix check is the proposed "successful use on a short safe prefix" promoted to a hard gate; it also lets the consensus threshold relax (recovering coverage) by catching the lone wrong anchors a consensus count cannot. Calibration was frozen on dev families (5001+); test seeds (100+) were never touched.

External substrate (VMAS physics). To test whether the effect depends on the authors' own grid and synthetic-continuous substrates, we transplanted not the diagnostic logger but the *whole runtime* (formation, certificate exchange, quarantine, receiver-side admission, frame-free continuous alignment, safe-prefix consumption) onto VMAS, a third-party continuous-physics multi-agent simulator. Holonomic agents move under a proportional controller in a landmark-populated box; a receiver with no shared frame recovers a witnessed water target purely by matching landmark constellations. Against a memory-free baseline given a dense box-tiling search and a longer horizon — so it *can* find the target by exhaustion, making the contrast a genuine speed-up rather than a floor — the full runtime reaches the target in 254.5 versus 543.5 team steps (-53.2%, $12/12$ paired seeds, sign test $p = 0.0005$), with transport fail-open $\mathbf{0.0000}$: agents that cannot anchor *refuse* and fall back to the same blind search rather than committing a phantom. Only the substrate density (landmark count, witness-sampling rate, consensus threshold matched to the constellation geometry) was recalibrated; every mechanism flag is unchanged from the grid configuration. The effect and its fail-closed safety therefore transfer to a third, externally implemented substrate.

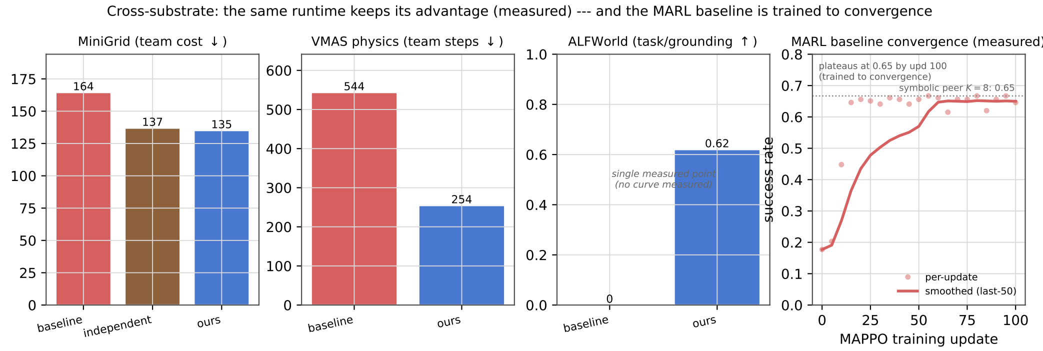

Figure 6: The integrated runtime keeps its advantage across substrates (measured summaries): MiniGrid team cost and VMAS physics team steps (lower is better), and ALFWorld task grounding (single measured point; no convergence curve was measured). The fourth panel is the measured MAPPO baseline training curve, which plateaus at success ≈ 0.65 — the symbolic-peer $K{=}8$ level — confirming the multi-agent-RL baseline is trained to convergence, so the runtime's advantage is not an artefact of an undertrained baseline. MAPPO baseline from the companion Q/R/M/C study (Anonymous, 2026a).

18 Related work

Zou et al. (2026) argue memory should preserve decision-relevant differences rather than descriptive similarity and gate merging by decision distortion; we adopt the intuition and add the coordination side: heterogeneous receivers, per-role utility profiles, and certificates recomputed by the receiver. Generative Agents (Park et al., 2023) score recency, importance and relevance — our importance is a measured counterfactual, and the U7 comparison quantifies the gap. MemRL (Zhang et al., 2026a) learns a utility of *using* stored episodes; Mem-α (Wang et al., 2025) learns construction operations with RL on question-answering reward — our controller learns *formation* under coordination reward with corruption probes. MemGAS (Xu et al., 2025) selects granularity; TrustMem (Yang et al., 2026) verifies consolidation transitions (coverage, preservation, faithfulness) — our exception operation and immutable episodic layer attack the same failure from the mechanism side, and our learned controller rediscovers append-not-overwrite from reward alone. Zhang et al. (2026b) show continuous rewriting eventually corrupts; our U1 measures the same corruption in the similarity baseline and quantifies the benefit of the exception mechanism. The present system combines structural analogy, causal utility, heterogeneous embodiments, coordination consequences, causally promoted landmark status and immutable provenance in one measured loop.

19 Limitations and open problems

Evidence classes. As in the companion studies, results are marked by causal strength rather than presented uniformly: the cadence bottleneck shift is the companion study's contribution, restated as background; CSM dominance and the equal-budget runtime comparisons are *controlled system comparisons*; the formation matrix (U1) is a *deterministic replay measurement*; the bandit and estimator results are *held-out learning comparisons*; the S1→S2→S3 social ladder is a sequence of *registered paired interventions*; the LLM active-context result is a *systems comparison* against raw context and five stronger context-management baselines (Appendix C, Axis 4) — it certifies the value of structured consolidation as such, with graph-specific value only on cross-frame identity, and is not evidence of a fundamental long-context limitation; and the order-theoretic and coalgebraic accounts are *interpretations*, with no empirical weight.

The present results establish an integrated prototype, but several limitations remain. First, the utility estimator is trained and validated in environments where exact paired replay is available for calibration. External stochastic environments require additional causal or off-policy validation. Second, the graph-matching analogy represents route structure but does not yet include a complete model of roles, affordances, preconditions, and causal effects. Third, landmark promotion is causal only within a predefined feature superset; open-ended proposal and naming of new concepts from learned representations remain unresolved. Fourth, the cross-family codebook captures repeated motifs but does not yet induce higher-order schemas with variables and role slots. Fifth, the social LLM experiments address active-context reconstruction rather than the complete multi-agent admission process. Sixth, the continuous and VMAS studies remain simulated and do not establish robustness under real sensors, actuation uncertainty, or robot hardware. Finally, the evaluation criterion is task and coordination utility under the implemented contract; it should not be interpreted as direct verification of the truth of every stored proposition.

The minimal CSM is evaluated primarily in controlled semantic navigation environments. The merge, trust, and staleness rules are explicitly specified rather than learned, and their transfer to richer perceptual or social settings remains to be established. The semantic place-correspondence mechanism assumes that the candidate vocabulary contains sufficiently discriminative local features; open symbol creation and heterogeneous sensor grounding are not solved in this paper. The categorical results formalize coherent gluing in $\mathbf{Pos}$ and should be interpreted accordingly. In particular, a colimit exists for disconnected diagrams; the phase analysis concerns whether one accepted correspondence component spans all agents, not whether a colimit exists at all. The coalgebraic section is an interpretive model of continuing behavior rather than a new control algorithm. Finally, the LLM context comparison uses two model families and controlled tasks. It shows that structured consolidation can resolve specific freshness and identity failures, but it does not establish general superiority over all long-context or production memory architectures.

20 Conclusion

This paper treated long-term collective memory as a distributed formation-and-admission process. Its first half established the minimal mechanics — merge, trust, staleness, and private-frame alignment — under which a distributed memory acts as an admissibility gate rather than a shared archive, and showed that the benefit comes from selective consumption, not communication volume. Its second half introduced a two-axis model of long-term memory formation in which counterfactual utility determines whether an observation should be retained and structural analogy determines how it should be integrated. The resulting controller distinguishes merging, exception storage, new-schema creation, repetition, and candidate rejection (dropping). Deterministic experiments show that the exception operation prevents the corruption caused by similarity-based overwriting, while learned and evolutionary controllers recover interpretable design principles, including append rather than overwrite and a finite route-description budget.

The long-horizon evaluation separates the functions of the main components. Structural consolidation reduces memory cost, but freshness and trust-aware consumption are required to preserve task performance under drift. Unregulated social exchange is harmful at long horizons; a world-time contract is necessary but insufficient; and receiver-side admission based on locally demonstrated utility changes the outcome in favor of collective exchange. The integrated runtime combines these mechanisms with provenance, private-frame alignment, route execution, reservation, rupture detection, and rollback. Under the tested equal-budget conditions, it outperforms raw history and independent memory, and the safety gates reduce fail-open behavior below the predefined threshold on the controlled substrates.

The current result is therefore an end-to-end research prototype of utility-gated, provenance-aware collective memory within the evaluated controlled substrates. It should not be interpreted as a universal solution to long-context memory for all LLM or robotic systems. Open-ended concept formation, richer causal analogy, broad social LLM evaluation, and real-world robotic validation remain the principal extensions.

A Categorical BDI reading of collective semantic memory

This section gives a compact theoretical reading of the water-search example in terms of belief, desire, and intention. The goal is not to introduce a new planner, but to make explicit the mathematical structure already used by the collective semantic memory architecture.

We assume a finite set of agents $A = \{1, \ldots, N\}$, a shared spatial basis X, and a shared semantic alphabet Σ. In the present grid-world implementation, X is the common coordinate frame and Σ contains tags such as `water_source`, `water_candidate`, `hazard_edge`, and `open`. This is an explicit modelling assumption. The agents may have different observations, different memories, and different confidence values, but they interpret places in a common coordinate framework and express semantic evidence in a common tag alphabet. Without the first assumption, peer merge requires a place-alignment problem to be solved first — Section 6 solves it constructively (fingerprint consensus recovers the inter-agent frame), and Section A.1 states what that construction is, categorically. Symbol alignment (a shared Σ) remains assumed.

What parameterizes commonality of interpretation. It is worth stating explicitly that the categorical setup localizes "how much the agents mean the same thing" in exactly three parameters with three different statuses. *(i) The alphabet* Σ — what counts as the same *predicate* — is a parameter of the category itself (the subscript of $\mathbf{Ctx}_\Sigma$) and is here *assumed*; relaxing it means constructing a translation between $\mathbf{Ctx}_{\Sigma_1}$ and $\mathbf{Ctx}_{\Sigma_2}$, which we leave open. *(ii) The alignment morphisms* of the merge diagram — what counts as the same *place* — were assumed in earlier drafts and are now *constructed* (Sections 6 and A.1), with their existence an empirically decidable predicate. *(iii) The trust$\times$staleness enrichment* — how much a peer's interpretation *counts* — makes commonality graded rather than binary, with its own closed-form horizon and a discrete trust-flip. The three are ordered by hardness: weights are tunable, place correspondence is solvable from co-observation, and predicate correspondence is genuinely open. The same triple reappears one level up, at the evaluator: scoring retrieval and materialization presupposes a correspondence between memory entities and world referents (see the companion Q/R/M/C study's limitations (Anonymous, 2026a)), and grounding *that* correspondence by the mechanisms of Section 6 is future work.

Parameter (i) has, in addition, a measurable *interior*: within a fixed Σ, which tag classes actually carry place correspondence is an empirical property, not a modelling choice. A registered vocabulary ablation in the companion route-transfer study (Anonymous, 2026b) makes the alphabet itself the treatment and finds the identity machinery *fail-safe in* Σ: impoverishing the vocabulary down to a single content class converts frame recovery into refusal, never into a wrong alignment (zero fail-open transports across 252 cells), and the consensus mass is attributable per class (hazards 0.56, walls 0.44, world borders 0.00). What no mechanism yet does is *select* Σ: here features receive landmark status from design plus the world's cooperation. A rule by which features *earn* that status through participation in successful alignments — the alphabet shaped by coordination rather than assumed — is the sharpest open end of parameter (i), and the natural meeting point with the symbol-alignment program of §A.4.

At time t, agent i maintains a local semantic context $\mathcal{K}_t^i = (G_t^i, \Sigma, I_t^i)$, where G_t^i is the set of place records stored in the agent's private memory, and

$$I_t^i \subseteq G_t^i \times \Sigma$$

is the thresholded incidence relation saying that place $g \in G_t^i$ carries tag $\sigma \in \Sigma$ with sufficient confidence. The graded implementation uses tag confidences and weighted scores; the crisp relation I_t^i is the thresholded skeleton needed for the theoretical description.

This formal context induces a Galois connection between sets of tags and sets of places:

$$\mathrm{ext}_t^i(B) = \{g \in G_t^i : \forall \sigma \in B,\ (g, \sigma) \in I_t^i\}, \qquad \mathrm{int}_t^i(S) = \{\sigma \in \Sigma : \forall g \in S,\ (g, \sigma) \in I_t^i\}.$$

For a water-search task, the desired semantic predicate is $B_{\mathrm{water}} = \{$`water_source`$\}$, or, in the implemented weighted form, a predicate combining positive evidence for `water_source`, `water_candidate`, and `water_nearby` with penalties for `hazard_edge` or nearby risk. Retrieval succeeds for agent i at time t exactly when

$$\mathrm{ext}_t^i(B_{\mathrm{water}}) \neq \varnothing.$$

Thus the retrieval stage is not a coordinate lookup. It is the test that the queried semantic meaning already has a non-empty extent in the agent's realized memory.

We can now define the BDI components.

Belief. The belief state of agent i is the structured semantic memory induced by $\mathcal{K}_t^i$. Equivalently, it is the concept lattice $L_t^i = \mathrm{Concepts}(\mathcal{K}_t^i)$, whose nodes pair extents of places with intensions of tags. A water belief is not the proposition "there is water at coordinate x." It is the region of L_t^i whose intension contains the water predicate. If this region is empty, the agent does not merely choose badly; the relevant semantic object is absent from its current memory.

Desire. Desire is a state-conditioned semantic predicate over the belief lattice. In the water example, the agent's internal state contains a thirst variable. The active desire can be written as

$$D_t^i = \begin{cases} B_{\mathrm{water}}, & \text{if } \mathrm{thirst}_t^i \geq \theta_{\mathrm{on}}, \\ B_{\mathrm{goal}}, & \text{otherwise.} \end{cases}$$

Thus desire is not an additional metaphysical object. It is the rule that selects which semantic predicate should be queried from the current belief structure.

Intention. Intention is the commitment to one materialized candidate. Once desire has selected a predicate and retrieval has produced a non-empty extent, the planner applies a satisficing choice function

$$\chi_t^i : \mathcal{P}(G_t^i) \to G_t^i \cup \{\perp\}, \qquad g_t^{i,\star} = \chi_t^i\big(\mathrm{ext}_t^i(D_t^i)\big),$$

where $\perp$ denotes failure to materialize a target. In Q/R/M/C terms, desire determines the query $Q^\star$, belief supports retrieval $R^\star$, and intention corresponds to materialization $M^\star$. Completion $C^\star$ is downstream of this semantic structure: it depends on whether the physical controller reaches the locked target and whether the target remains available under multi-agent occupancy.

This gives a compact categorical reading. Let $\mathbf{Ctx}_\Sigma$ be the category whose objects are semantic contexts (G, Σ, I) over the fixed alphabet Σ, and whose morphisms preserve semantic incidence. A local memory trajectory for agent i is a time-indexed diagram

$$\mathcal{K}_0^i \to \mathcal{K}_1^i \to \cdots \to \mathcal{K}_t^i \to \mathcal{K}_{t+1}^i \to \cdots$$

in $\mathbf{Ctx}_\Sigma$. Each arrow is induced by observation and memory update. When new evidence is consolidated, the formal context changes; therefore its concept lattice changes. This is the order-theoretic content of memory growth.

In Q/R/M/C terms, an empty extent $\mathrm{ext}_t^i(D_t^i) = \varnothing$ is a belief/retrieval failure, a non-empty extent with no stable lock is a materialization failure, and a successful lock that cannot be reached is a completion failure — the semantic and the physical parts of the agent's state fail in different, separately measurable ways.

The collective case lifts this by one construction. At broadcast times the receiver i holds a diagram of local memory objects $\mathcal{D}_t^i = (\mathcal{K}_t^i, \mathcal{K}_t^{j_1}, \ldots, \mathcal{K}_t^{j_m})$ together with alignment maps, and queries the merged consensus object $\widetilde{\mathcal{K}}_t^i = \mathrm{Merge}_{\mathrm{trust,stale}}(\mathcal{D}_t^i)$, with peer evidence weighted exactly by the rules of Section 5 ($\mathrm{trust}_{j\to i} \cdot e^{-\alpha\,\mathrm{age}_j}$). The same BDI definitions apply to the merged context: intention is now selected from $\mathrm{ext}_{\widetilde{\mathcal{K}}_t^i}(D_t^i)$ rather than from the private extent alone.

This explains why communication cadence produces a bottleneck shift. Faster broadcast increases the freshness of peer evidence. This tends to enlarge or improve the water extent $\mathrm{ext}_{\widetilde{\mathcal{K}}_t^i}(B_{\mathrm{water}})$, so agents can materialize targets more reliably; in Q/R/M/C terms, this raises $P(M^\star \mid R^\star)$. But the same fresh shared belief also synchronizes intentions. Several agents may lock onto the same scarce water source. The memory is semantically accurate, but the physical target is now contested. This lowers $P(C^\star \mid M^\star)$. The bottleneck is therefore physically grounded. It is not a formal artifact of the decomposition: it arises because semantic belief is shared through memory, while completion is constrained by occupancy and scarce resources. Collective semantic memory must therefore regulate not only how much information is shared, but also how it is merged, whose evidence is trusted, and when stale evidence should stop influencing intention formation.

In this sense the "collective" is not a central mind and not a global map but a stabilised family of peer-indexed processes; the next subsection makes that statement precise.

The dual, process-level reading — agents as coalgebras (codata), the collective as a final coalgebra, communication as depth-one corecursion, and the bottleneck shift as a competition between belief-and intention-bisimilarity — is developed in Appendix B. It is a lens, not a mechanism: nothing downstream depends on it, which is why it lives in the appendix.

A.1 Colimit merge and its invariants

The merge object above can be described more precisely. Working in a timestamp- and provenance-enriched version of $\mathbf{Ctx}_\Sigma$, a snapshot merge from agents $1, \dots, N$ is a diagram of local memory objects plus alignment maps induced by the place-identity relation. The consensus object $\widetilde{\mathcal{K}}_t^i$ is the colimit of this diagram, implemented concretely by the union-find merge over aligned concepts. This reading gives three useful constraints.

1. **Order independence.** The merged object is unique up to isomorphism, so the result is not an artefact of processing agent snapshots in a particular order.
2. **Chaining is expected.** If concept A aligns with B and B aligns with C, union-find performs the transitive closure. Over-merging is therefore not a bug in the data structure; it is the cost of the chosen equivalence relation.
3. **Decay is not a morphism.** Staleness is kept as a weight field outside the alignment morphisms. Otherwise age-then-merge and merge-then-age need not commute, breaking the clean interpretation of broadcast snapshots as one diagram.

The colimit reading requires the alignment maps of the diagram; rather than assume them, Section 6 constructs them, which gives the colimit reading its scientific function:

1. **The alignment morphisms are constructed, not assumed.** Fingerprint consensus IS an algorithm that produces the alignment maps of the diagram — mutually-unique landmark matches determine the frame transform through which every local memory object embeds.
2. **Faithful consensus becomes an empirical predicate.** A unique faithful alignment edge is instantiated only when semantic consensus is sufficiently discriminative. The colimit itself still exists for an unaligned or disconnected diagram (§A.2: $\mathbf{Pos}$ is cocomplete), but it then represents separated components — a coproduct of sub-consensuses — rather than a single consensus-spanning memory. In a $180°$-symmetric world the available observations do not identify a unique admissible alignment morphism (two incomparable diagrams support equal consensus), and the runtime therefore refuses to instantiate the edge — the fail-closed behavior of Section 6 is exactly this non-instantiation made operational. The predicate is measured, not asserted: the round-trip defect of §6 is exactly 0 when the recovered transformations are mutual inverses on the observed substructure within the implemented discrete transformation family (in every recovered translation and $90°$-rotation case, $30/30$), and the edge is left undefined when the observations do not identify a unique admissible alignment (fail-closed, $30/30$, with zero spurious gluings); a scalar witness for whether the colimit is consensus-spanning rather than fragmented.
3. **Decay stays outside the morphisms** (invariant 3), and empirically the trust$\times$staleness gate is unchanged by frame recovery — the same integer breakpoints with and without a shared frame — confirming that the two layers commute as the diagram reading requires.

A.2 Collective memory as a coherent gluing of local place categories

The colimit merge of §A.1 *assumed* the alignment maps and §6 *constructs* them; we can now state, as the central categorical result, what object the collective memory is when agents hold *different* frames, and exactly when it exists faithfully.

Setup. Agent i's realized memory is a thin category (a poset): its concept lattice $L_i = \mathrm{Concepts}(\mathcal{K}_i)$ over i's private frame, objects the places, order the extent inclusion of §A. Frame recovery (§6) yields, on the co-observed subposet, a monotone map $F_{ij} : L_i \to L_j$ that transports i's places into j's frame, and its reverse F_{ji}. Because the categories are *thin*, a recovered pair (F_{ij}, F_{ji}) is an *adjoint pair* precisely when it is a Galois connection, with unit $\eta^{ij} : \mathrm{Id}_{L_i} \Rightarrow F_{ji} \circ F_{ij}$ — the adjunction defect measured in §6.

Definition (coherent alignment system). The family $\{F_{ij}\}$ over the communication graph is *coherent* if (i) each pair is an adjoint order-embedding (η^{ij} an isomorphism, i.e. defect 0: no place is aliased onto a distinct one), and (ii) it satisfies the cocycle condition $F_{jk} \circ F_{ij} = F_{ik}$ on triple overlaps.

Proposition (two agents). For an adjoint pair $F \dashv G$ between L_A, L_B, the collage (comma) category $\mathrm{Gl}(F)$ — objects $L_A \sqcup L_B$, cross-hom $\mathrm{Gl}(F)(X, Y) = L_B(FX, Y) \cong L_A(X, GY)$ (Mac Lane, 1998) — contains L_A, L_B as full subposets and is the two-agent collective memory; the defect η_X is the obstruction to X round-tripping to itself, and the gluing collapses distinct places iff η is not an isomorphism.

Theorem (collective memory = colimit of the alignment diagram). Let $\mathbf{D}$ be the diagram in $\mathbf{Pos}$ (equivalently the enriched $\mathbf{Ctx}_\Sigma$) whose vertices are the L_i and whose edges are the F_{ij}. The collective memory is $\operatorname{colim} \mathbf{D}$. As $\mathbf{Pos}$ is cocomplete this colimit always exists; it is a *faithful* gluing — every L_i embeds and no two distinct places are identified — *iff* the alignment system is coherent. When all $F_{ij} = \mathrm{id}$ (a shared frame) $\mathbf{D}$ is the constant diagram and $\operatorname{colim} \mathbf{D}$ is exactly the union-find colimit merge of §A.1; the general statement removes the shared-frame assumption by replacing identities with the recovered order-embeddings (mutual inverses on the observed substructure).

Argument. Cocompleteness of $\mathbf{Pos}$ realizes the colimit as the disjoint union quotiented by the preorder generated by the F_{ij}-identifications. Each canonical injection $L_i \hookrightarrow \operatorname{colim} \mathbf{D}$ is faithful iff those identifications add no collapse inside an L_i, which holds iff every F_{ij} is an order-embedding (unit iso, defect 0) and the identifications agree across overlaps (cocycle) — i.e. coherence. The shared-frame case is the constant diagram, whose colimit is the componentwise union, i.e. union-find. $\square$

Corollary (two computable witnesses). Coherence is checkable without solving general semantic alignment. (a) The pairwise *round-trip defect* η (§6) is zero iff each recovered edge is a mutual inverse on the observed substructure within the discrete transformation family — measured exactly 0 in $30/30$ recovered translation and rotation cases — and guards against *over-merge*. (b) The *connectedness* of the trust×cadence reachability graph (§A.3, Figure 7) decides whether a single $\operatorname{colim} \mathbf{D}$ spans all N or the diagram disconnects into a coproduct of sub-consensuses (*fragmentation*). The two failure modes are categorically distinct and separately measured: defect > 0 is collapse, disconnection is fragmentation — the two axes of Figure 7 and the fail-closed behavior of §6.

Scope. The proofs are the standard cocompleteness of $\mathbf{Pos}$ and the collage construction (Mac Lane, 1998); the theorem's job is to name the object and pin its existence to two measured quantities. The empirical content is that fingerprint recovery yields mutually inverse order-embeddings on the observed substructure, thereby instantiating the coherent-alignment case used by the categorical model; it lives in §6, not in the theorem. One assumption is *not* discharged: a shared tag alphabet Σ; with different alphabets the F_{ij} would have to translate Σ as well, which is beyond the present correspondence mechanism and remains unresolved.

A.3 Protocol parameters as categorical structure: a trust×cadence phase diagram

The reading so far models the *memory* (objects are semantic contexts, merge is their colimit) under a shared frame and full exchange. Its predictive use comes from turning the lens on the *communication structure* and asking *what shape the colimit takes* as a function of the two protocol parameters — so that the category computes a structural fact rather than re-describing an algorithm. (In $\mathbf{Pos}$ the colimit of the diagram always *exists* by cocompleteness, disconnected or not; the structural question is whether it is a single consensus component or a coproduct of sub-consensuses, exactly the dichotomy of §A.2.) Two parameters play two structural roles:

- **Trust τ gates morphisms.** An edge $i \to j$ (agent i incorporates j's evidence) exists only where i's trust in j is at least τ. Raising τ prunes the trust graph.
- **Cadence K gates composition depth.** Over a horizon H there are $R = \lfloor H/K \rfloor$ broadcast rounds, so a composed chain $i \to k \to \cdots \to j$ can be at most R morphisms long. Low K (many rounds) allows deep composition; high K allows only short paths.

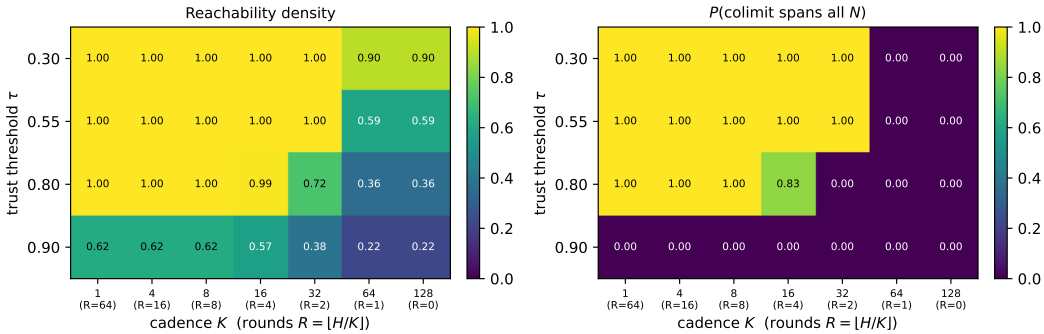

Figure 7: Communication-structure phase diagram over trust threshold τ (rows) and cadence K (columns; $R = \lfloor H/K \rfloor$ broadcast rounds), N=8 agents on an interaction ring, 12 seeds. **Left:** reachability density (fraction of ordered agent pairs whose evidence can compose within R rounds). **Right:** P(the colimit spans all N), i.e. all-to-all reachability. The diagonal boundary is a structural transition: the collective admits a single consensus object in the low-τ/low-K region and fragments in the high-τ/high-K corner, with higher τ demanding lower K (more rounds) to stay connected.

A *consensus-spanning* colimit — a single connected consensus component that fuses and serves every agent's evidence — obtains iff the R-round reachability of the τ-thinned trust graph is all-to-all; otherwise the colimit degenerates into a coproduct of sub-consensuses.

Figure 7 sweeps $\tau \times K$ for N=8 agents whose pairwise trust is produced by the system's own EMA rule (`peer_memory.peer_trust.AsymmetricTrust`) under a ring co-observation topology (neighbours co-observe often and earn high trust; distant agents do not), averaged over 12 seeds (`experiments/collective_semantic_memory/phase_diagram_trust_cadence.py`). The result is a genuine phase transition with a diagonal boundary. At low τ the trust graph is dense and the colimit spans all agents at every cadence. Raising τ prunes the long-range morphisms down to a near-neighbour ring; the collective then keeps a spanning colimit *only if* cadence is low enough for multi-hop composition through still-trusted intermediaries to reconnect it — τ=0.55 holds the spanning colimit down to R=2, τ=0.80 needs $R \geq 4$ (the ring diameter), and τ=0.90 fragments throughout. Cadence and trust are therefore not opaque knobs: together they determine a categorical structural invariant, and the phase boundary is exactly where the collective transitions from a single consensus object to fragmented sub-consensuses.

This is the sense in which the categorical model is operational rather than descriptive: each protocol parameter maps to a concrete change in the communication diagram, and the diagram's structural invariant (whether the colimit spans all agents or splits into a coproduct) predicts a qualitative regime of the collective. *Scope.* The trust matrix is produced by the real EMA rule under a modelled ring co-observation topology, and "consensus-spanning" is operationalized as all-to-all R-round reachability; establishing quantitatively that crossing this structural boundary predicts the drop in observed Q/R/M/C performance is the natural next experiment.

A.4 Toward symbol alignment: grounding tag alphabets through shared places

The gluing theorem (§A.2) still assumes a shared alphabet Σ. Dropping it, each agent holds a context (G_i, Σ_i, I_i) over its *own* vocabulary, and the right morphism is no longer a functor over one Σ but an *infomorphism* (Barwise and Seligman, 1997): a pair $f : G_A \to G_B$, $\hat{f} : \Sigma_B \to \Sigma_A$ with $I_B(fg,\, y) \Leftrightarrow I_A(g,\, \hat{f}y)$. Infomorphisms compose and have colimits (Barwise–Seligman channels), so the collective memory over heterogeneous alphabets is the *channel core* — the colimit of the infomorphism diagram — which extends §A.2 verbatim (functors over one Σ become infomorphisms over the Σ_i; shared Σ with $\hat{f} = \mathrm{id}$ is the old case).

Grounding mechanism. Symbols align through shared referents: on the places already anchored by frame recovery (§6), the co-occurrence of an A-tag and a B-tag is evidence they name the same feature, and $\hat{f}$ is read off the anchored co-occurrence. (When the alphabets differ so much that fingerprints themselves cannot seed the anchors, one bootstraps from the alphabet-independent structural constellations and alternates recovery of f and $\hat{f}$.) The *symbol-alignment defect* def_Σ — the fraction of anchored (place, y) pairs violating the infomorphism biconditional — is the symbol analogue of the frame adjunction defect of §6.

Preliminary result. A controlled study (`experiments/warp/symbol_alignment.py`; $|\Sigma_A|$=8, 120 places, 20 seeds) gives agent B a secret relabelling of A's alphabet with observation noise, and recovers $\hat{f}$ from anchored co-occurrence.

Table 3: Symbol alignment by grounding vs. assuming identical alphabets. Target-ID F1 is identifying A's target feature from B's (translated) evidence. "naive" assumes the alphabets coincide by index; "recovered" learns $\hat{f}$; "oracle" is the true map.

noise	transl. acc.	def_Σ	F1 naive	F1 recovered	F1 oracle
0.00	1.00	0.000	0.56	1.00	1.00
0.10	1.00	0.099	0.55	0.92	0.92
0.30	0.89	0.304	0.56	0.69	0.73

Assuming identical alphabets fails (F1 ≈ 0.55: the relabelling sends B's target tag to the wrong feature); grounding recovers the map (translation accuracy 1.00, def_Σ tracking the noise floor) and reaches oracle-level target identification, robustly down to 15% anchored places (accuracy 0.95). Coarsening two features into one B-tag bounds the achievable accuracy — there is no faithful bijection, so the channel core coarsens rather than errs. Symbol alignment is thus not a wholly open problem but the same grounding-plus-defect recipe one alphabet up; a full treatment (genuine incommensurability, online drift) is the next paper.

B The coalgebraic layer: individuals as codata, the collective as a final coalgebra

The reading of §A fixes belief *content*: at a state, the concept lattice $L^i_t = \mathrm{Concepts}(\mathcal{K}^i_t)$ says what is retrievable, and the merge of §A.1 says how peer contents combine into $\widetilde{\mathcal{K}}^i_t$. Both are *algebraic*: they build a belief object. Neither speaks to the agent as a non-terminating *process* that must emit an action and a signal at every tick under a moving environment, nor to the sense in which a *population* comes to share a convention. That process side is *coalgebraic* (Rutten, 2000; Jacobs, 2016). We make it precise on the water-search example and show the two sides are dual, meeting inside a single state transition. (The belief/desire/intention vocabulary follows the BDI agent tradition (Rao and Georgeff, 1995), here given coalgebraic content.)

Signature. Fix an action set Act (move, dig, noop), a per-agent observation set O_i (agent i's local percept — its *unique view*), and an *ostension* set

$$M \;=\; 1 + (X \times \Sigma),$$

whose values are either *silence* (the summand 1) or a pointer "place x carries tag σ" (whistle-and-point at a water-marked cell). That silence is a first-class value of M matters: the ostension channel *can be informatively empty*, and it is separate from the state update below. This is exactly the observability that a learned real-valued message vector destroys (its channel is never empty and never separates signalling from stepping), and it is why the symbolic stages remain readable.

State. We take the coalgebra carrier to be the H-MDP high-level state of §A,

$$\mathcal{S}_i \;=\; \big\{\, h = (\mathcal{K}^i,\, D^i,\, g^{i,\star},\, s^i) \,\big\},$$

carrying the same information as $(L^i, D^i, g^{i,\star}, s^i)$ since $L^i = \mathrm{Concepts}(\mathcal{K}^i)$. Thus *belief lives in the state*; the lattice is a projection of it.

Definition (codata: the agent's cotype). The observable behavior of agent i is a coalgebra for the endofunctor

$$B_i(S) \;=\; \underbrace{\mathrm{Act}}_{\text{act now}} \;\times\; \underbrace{M}_{\text{signal now}} \;\times\; \underbrace{S^{\,O_i \times M^{A\setminus\{i\}}}}_{\text{continue given (my next percept, peers' signals)}},$$

i.e. a map $c_i = \big\langle\, \texttt{observe\_act}_i,\ \texttt{observe\_ostension}_i,\ \texttt{step\_intension}_i \,\big\rangle : \mathcal{S}_i \to B_i(\mathcal{S}_i)$ with three destructors

$$\texttt{observe\_act}_i : \mathcal{S}_i \to \mathrm{Act}, \quad \texttt{observe\_ostension}_i : \mathcal{S}_i \to M, \quad \texttt{step\_intension}_i : \mathcal{S}_i \to \mathcal{S}_i^{O_i \times M^{A\setminus\{i\}}}.$$

Here $M^{A\setminus\{i\}} = \prod_{j \neq i} M$ is the tuple of peers' ostensions delivered this tick. Operationally, $\texttt{step\_intension}_i(h)(o, (m_j)_{j\neq i})$ (i) folds the percept o and the peer pointers (m_j) into the context by the trust/staleness merge, $\widetilde{\mathcal{K}}^i_t = \mathrm{Merge}_{\text{trust,stale}}(\cdots)$ of §A.1; (ii) recomputes the lattice; (iii) re-evaluates the desire predicate D^i; and (iv) re-locks the intention $g^{i,\star} = \chi^i_t(\mathrm{ext}(D^i))$ of §A. So Merge and χ are *components of one destructor*. The agent *is* the codata generated by c_i: it has no finite terminal plan, only the standing ability to emit an action and an ostension and to continue. This is the “infinite survival generator” of the motivating example.

Definition (indexed family). The system is the A-indexed family $\{(\mathcal{S}_i, c_i)\}_{i\in A}$, a coalgebra in $\mathbf{Set}^A$. The functor B_i depends on the index i through O_i (and through the peer set $A \setminus \{i\}$): the same survival cotype is instantiated at each agent through that agent's local viewpoint. Agent 1 on the dune generates from “I see the horizon”; agent 2 in the hollow generates from “I feel the soil.” The index is the viewpoint.

Definition (codependence: the collective coalgebra). Let the environment deliver a joint percept $o = (o_i)_{i\in A} \in \prod_i O_i$. The collective one-tick map on $\prod_i \mathcal{S}_i$ wires each agent's ostension output into the others' step inputs,

$$\Phi\big((h_i)_i\big)(o) \;=\; \Big(\, \texttt{step\_intension}_i(h_i)\big(o_i,\ (\texttt{observe\_ostension}_j(h_j))_{j\neq i}\big) \,\Big)_{i\in A}.$$

Because i's next state is a function of the *destructor outputs* of the others, the family is not N independent processes but one coalgebra for the product functor $\mathbf{B}(S) = \big(\prod_i(\mathrm{Act} \times M)\big) \times S^{\prod_i O_i}$ on $\prod_i \mathcal{S}_i$. This wiring is the *codependence*: agent 1 cannot step without passing agent 2's ostension through its own `step_intension`. Broadcast cadence enters here as a gate: at non-broadcast ticks the delivered tuple (m_j) is the last received (stale) tuple, or silence; K is the delivery period.

Definition (behavior and bisimulation). B_i is a container (polynomial) functor, so it has a final coalgebra $(\nu B_i,\ \mathrm{beh}_i)$ (Aczel, 1988; Rutten, 2000) whose elements are the $\big(O_i \times M^{A\setminus\{i\}}\big)$-branching, $(\mathrm{Act} \times M)$-labelled behavior trees, and there is a unique coalgebra morphism $!_i : \mathcal{S}_i \to \nu B_i$ sending a state to its whole future behavior. A B_i-*bisimulation* is a relation $R \subseteq \mathcal{S}_i \times \mathcal{S}_i$ under which related states emit equal action and equal ostension and, on every input, step to R-related states. Container functors preserve weak pullbacks, so the *greatest bisimulation* $\sim_i$ exists, is an equivalence, and equals $\ker(!_i)$: two states are bisimilar iff they have the same behavior (Sangiorgi, 2011). The same holds for the collective coalgebra $\big(\prod_i \mathcal{S}_i,\ \langle(\texttt{observe\_act}_i)_i, (\texttt{observe\_ostension}_i)_i, \Phi\rangle\big)$, with greatest bisimulation $\approx$.

Definition (society / culture). The *collective* is the image of the states in the final coalgebra $\nu\mathbf{B}$, equivalently the quotient $\big(\prod_i \mathcal{S}_i\big)/\approx$. It is neither a central server nor a shared map: it is a bisimilarity class of joint behavior. A *water-search convention* is this invariant read on the water sub-signature. Let $M_{\text{water}} \subseteq M$ be the pointers whose tag lies in the water family, and let $\nu B_i \to \nu B^{\mathrm{w}}$ be the forgetful map to the behavior trees restricted to receiving M_{water} and emitting the corresponding intention lock. The population *holds the convention* when all agents' reachable behaviors agree under this map: hearing “point-and-dig at a water-marked place” drives every agent's intention destructor to lock that place (up to occupancy). Governance is then automatic in both directions. A new agent (the example's fourth agent) is a member of the society exactly when its behavior lands in the class $[\cdot]_\approx$ on M_{water} — otherwise its ostensions are not understood and its own step never converges (top-down); and the class itself was reached by the local, coupled steps of the first agents at one green shrub (bottom-up).

Remark (communication = lazy, depth-one corecursion). The behavior $!_i(h)$ is defined by unfolding c_i indefinitely. On receiving $m_j = \texttt{observe\_ostension}_j(h_j)$, agent i uses only this *depth-one* destructor output inside $\texttt{step\_intension}_i$; it does *not* compute $!_j(h_j) \in \nu B_j$, the peer's full behavior, which is the “I think that you think…” regress. Communication is therefore corecursion

Table 4: The motivating example's terms as formal objects of the coalgebraic reading. Each row is the water-search meaning of a construction already used in §B.

Informal term	Formal object	Water-search meaning
Cotype (codata)	B_i-coalgebra $(\mathcal{S}_i, c_i)$	the agent's non-terminating action/signal generator
Indexed type	family over A; dependence of B_i on O_i	the agent's unique local viewpoint (dune vs. hollow)
Codependent type	wiring $\texttt{observe\_ostension}_j \to \texttt{step\_intension}_i$ in Φ	"I step only through your ostension"
Individual (instance)	a state $h \in \mathcal{S}_i$ and its behavior $!_i(h)$	the running agent digging and whistling now
Destructors (cogenerators)	`observe_act`, `observe_ostension`, `step_intension`	step-and-dig; whistle-and-point; fold neighbour's signal into next state
Lazy evaluation	depth-one corecursion + relevance stop	unfold the peer's coalgebra one step, not its whole behavior
Final coinductive core / society	$\nu\mathbf{B}$; greatest bisimulation $\approx$ (on M_{water})	the stable, distributed water-search culture

truncated to the least depth that resolves the receiver's own next step — a finite approximation of the final-coalgebra map — with a relevance-theoretic stopping rule (Sperber and Wilson, 1995) deciding that depth. The regress collapses into one effective action.

Remark (algebra–coalgebra bridge to §A–A.1). Belief is algebraic: the FCA lattice L_t^i built by the colimit merge is the *content at a state*. The agent is coalgebraic: the codata observed by the three destructors is the *process over states*. They are dual and meet in one place — the colimit merge and the satisficing choice χ are the internals of `step_intension`. The memory-growth diagram $\mathcal{K}_0^i \to \mathcal{K}_1^i \to \cdots$ of §A is precisely the belief-projection of the coalgebra trajectory $h_0^i \mapsto h_1^i \mapsto \cdots$.

Coalgebraic reading of the bottleneck shift. The result of §4 and Table 1 has a one-line reading here. Two kinds of bisimilarity compete. Bisimilarity on the *ostension / belief* sub-signature is the wanted invariant: it is the shared water convention. Bisimilarity on the raw *intention* output over a *scarce* target is the unwanted one: several states mapping to the same $g^\star$ is a collision. Occupancy $\mathcal{O}_{-i}$ is the extra-coalgebraic constraint that penalises the second. Cadence K tunes how hard the coupling Φ drives the collective toward the *full* greatest bisimulation $\approx$: small K delivers fresh ostensions often, enlarging the merged water extent so intention locks reliably ($P(M^\star \mid R^\star)\uparrow$), but over-synchronises the intention destructor so agents collide ($P(C^\star \mid M^\star)\downarrow$). The interior optimum $K{=}8$ is where belief converges without intention collapsing onto one cell; the minimal CSM adds trust and staleness precisely so that `step_intension` reaches belief-bisimilarity (fresh, trusted, aligned evidence) while damping intention-bisimilarity (stale evidence decays before it can herd every agent onto the same lock). We state this as a reading consistent with the data, not as a theorem: the container functors, their final coalgebras, and bisimulation are standard constructions used here as a lens, and the relevance-theoretic stopping rule is an interpretive gloss rather than a fitted mechanism.

C Structured memory versus raw context under a fixed model

Modern LLMs *accept* long context; the question this appendix measures is whether they turn it into a stable, globally coherent state — and whether the Songlines graph supplies that state instead. In every arm the *same* model answers; only the memory substrate differs: **raw** (the full observation transcript in the prompt — typical usage) vs. **graph** (the same evidence consolidated by the Songlines rule: latest state per place, staleness explicit, $O(\text{salient places})$ prompt). Three axes (`experiments/context_vs_structure/`; HF backend on SLURM GPU nodes and local Ollama as an independent second backend).

Axis 1: volume and integration (one witness log). A witness log over a $30{\times}24$ grid contains a freshness conflict (water seen repeatedly at site A, later dried; one fresher sighting at B). Questions: *fact* (where is water now), *count* (how many waters now), *hop* (nearest hazard to current water).

Table 5: Accuracy by question, arm, and log volume L (lines). Qwen2.5-7B (cluster, 20 seeds/cell) on cells within its compute envelope; Llama-3.1-8B (local, 10 seeds/cell) at large L with *verified full ingestion* (`prompt_eval_count`=25,030 tokens at L=2000; no truncation). Graph prompts stay ≈350 bytes at every L.

Question	Arm	L=50	L=200	L=800	L=2000
fact	raw	1.00 / –	1.00 / –	– / 0.00	– / 0.00
fact	graph	1.00 / –	1.00 / –	1.00 / 1.00	1.00 / 1.00
count	raw	**0.00** / –	**0.00** / –	– / **0.00**	– / **0.00**
count	graph	1.00 / –	1.00 / –	1.00 / 1.00	1.00 / 1.00
hop	raw	0.00 / –	0.05 / –	– / 0.00	– / 0.00
hop	graph	0.00 / –	0.15 / –	0.40 / 0.50	0.30 / 0.60

Two failure layers, both with identified mechanisms. *Cognitive:* the raw arm loses global state at **any** volume — on *count* it answers "2" (17/20) or "3" (3/20) when the truth is 1: it adds up historical mentions instead of resolving freshness, i.e. no consolidation. At large L even single-fact retrieval collapses despite verified full ingestion (a probe with the answer at the very end of a 25k-token prompt is also missed). *Systemic:* on the cluster's 24 GB sm75 GPUs, raw-arm inference at 5–10k tokens exhausts memory through the quadratic attention matrix (marked as compute-envelope cells, "–", never reported as cognition), while the graph arm runs at constant cost at every volume. Raw context loses twice — as cognition and as systems cost; *hop* identifies the empirical boundary: structure repairs retrieval, not arithmetic reasoning.

Axis 2: time (sessions). A persistent world drifts (water relocates, p=0.35/session); after each of 8 sessions the model is asked where the water is now, from **raw replay** (all past transcripts re-injected, the typical chat pattern) vs. the persisted **graph**. Qwen2.5-7B: the graph arm holds 0.62–1.00 across all eight sessions at a flat ≈0.3 kB prompt; the replay arm's valid cells are already at 0.00–0.67 in the earliest sessions and its prompt grows to 25 kB by session 8 (later cells envelope-limited on the cluster; on the local backend, where no envelope exists, replay degrades to 0.00–0.50 by 19 kB — the same pattern).

Axis 3: system size (multi-agent, private frames). Three agents accumulate observations over G sessions, each logging in its *own* secret coordinate frame. Question: "how many DISTINCT water sources exist?" — the same physical water seen by two agents at different private coordinates must be counted once. The raw arm gets all three logs (and is told the frames differ); the graph arm gets the Songlines machinery's merged map (fingerprint frame recovery, §6, plus dedup). Qwen2.5-7B: graph accuracy *rises* with growth — $0.40, 0.90, 1.00, 1.00$ at G=$1, 2, 4, 8$ (≈330 bytes) — because frame alignment locks as fingerprints accumulate (the exact% curve of the growth study); the raw arm (13k–105k chars) does not produce a single correct answer on any envelope-valid cell. Cross-frame identity is precisely what a context window cannot represent and the gluing structure can.

Axis 4: stronger baselines at a fixed token budget. The raw transcript is a weak opponent, so we re-ran the freshness and (single-log) identity questions against five stronger context-management baselines under one budget accounting (identical model, evidence stream, questions, and scoring; token counts by one tokenizer; payload $\leq$ budget enforced; artifact `tmp/llm_context_baselines/`; 12 layouts per cell, Llama-3.1-8B local): **raw+instructions** (transcript plus explicit conflict-resolution guidance, preamble charged to the budget), **rolling summary** (incremental LLM summarization; its extra calls are logged as build cost), **retrieval** (top-k TF–IDF chunks), **table** (a *programmatic* latest-state-per-place table, no LLM, no graph), and the Songlines **graph**.

Three honest readings. (i) The strongest baseline is the cheapest one: the programmatic latest-state *table* matches the graph on freshness at every budget (and on the single-log identity question), at 61–76 tokens. What these two question families certify is therefore the value of *external structured consolidation as such*, not of the graph representation specifically. (ii) The failure modes of the other baselines are informative: TF–IDF retrieval fails freshness at every budget (0.08–0.17) because lexical similarity preferentially returns the *stale* sightings; rolling summary is competitive on freshness but carries a hidden compute cost (144 extra LLM calls per log) and loses the identity question; instructions rescue the raw transcript at moderate budgets (1.00 at 192) but not at full length (0.42 uncapped) — the conflict-resolution failure is attentional, not informational. (iii) The graph's

Table 6: Fixed-budget comparison, Llama-3.1-8B, long logs (≈2,000-token transcripts), freshness / identity accuracy at representative budgets; “tok” = measured payload, “bc” = summary build calls. The uncapped column reproduces the published raw degradation (0.08).

	uncapped		budget 192		budget 96	
Arm	fresh	tok	fresh	tok	fresh	tok
raw	0.08	1979	0.58	182	0.58	87
raw+instr	0.42	2097	**1.00**	176	0.00	96
summary	0.83	97 (bc 144)	0.92	90	0.75	70
retrieval	0.08	2001	0.08	176	0.17	81
table	**1.00**	76	**1.00**	76	**1.00**	76
graph	**1.00**	117	**1.00**	117	0.33	58

differential value over a flat table does not live on these two axes: it lives on Axis 3, cross-frame identity, where a latest-state table in one agent’s private frame cannot express the merge at all — that is the axis that requires the alignment machinery. At the tightest budget (96) the graph serialization is truncated and degrades (0.33) while the more compact table survives; representation overhead is a real cost and is reported as such. The same seven-arm grid on a second model family (Qwen2.5-3B, cluster GPU, same budgets and scoring) reproduces the ordering: table and graph at the top of freshness at every budget, TF–IDF retrieval weakest (0.42–0.58 on long logs), and the flat table again matching or beating the graph on the single-log identity question — the structured-consolidation attribution is not model-specific.

Summary. The same model, given the same evidence, preserves long, drifting, multi-agent context when the evidence reaches it in consolidated structured form — the Songlines graph, or, for the single-frame freshness and identity questions, even a flat latest-state table (Axis 4): flat-cost prompts, freshness resolved. Given the raw stream instead, it fails on integration at 1.3 kB, on freshness at 19 kB with full ingestion, and on cross-frame identity at every tested size. The graph-specific value concentrates where flat structure cannot follow: cross-frame gluing (Axis 3). Scope: two model families, one task family per axis, and the compute-envelope cells are reported as such; the graph arm’s consolidation here is the deterministic Songlines rule — LLM-driven consolidation is the roadmap’s next stage.

D Reproducibility commands

- Main cadence sweep: `pythonexperiments/big_experiment/exp_cadence_phase.py--modefull--workers8--out_dirtmp/big_experiment_qrmc_40`
- Extended cadences: `pythonexperiments/big_experiment/exp_extra_K.py--workers8--out_dirtmp/big_experiment_extraK`
- Q/R/M/C analysis: `pythonexperiments/big_experiment/analyze_qrmc.py--runs_csvtmp/big_experiment_qrmc_40/runs.csv--out_dirtmp/big_experiment_qrmc_40`
- Minimal CSM benchmark: `python-mexperiments.collective_semantic_memory.run_csm_benchmark--out_dirtmp/csm_benchmark`
- MiniGrid portability sweep: `python-mexperiments.minigrid_multiagent_wrapper.run_portability_sweep--out_dirtmp/minigrid_portability`
- Adjunction round-trip defect of frame recovery (Section 6): `pythonexperiments/warp/alignment_defect.py`
- Context-vs-structure suite (Appendix C): `experiments/context_vs_structure/run_ctx_vs_struct.py`, `run_persistence_llm.py`, `run_growth_llm.py` (HF backend on GPU or –backend ollama locally)
- Trust×cadence structural phase diagram (Section A.3, Figure 7): `pythonexperiments/collective_semantic_memory/phase_diagram_trust_cadence.py--figfigures/fig_phase_diagram.pdf`

Every central result has a command (all in `experiments/song_grammar/`, run with `PYTHONPATH=.`):

```
# formation core
exp_u1_ucsm_smoke.py                         # U1 two-axis matrix
exp_u2_bandit.py    --train-seeds 0 24 --eval-seeds 100 110 --episodes 40
exp_u3_evolution.py --generations 30 --pop 16 --episodes 25
exp_e1_landmark_emergence.py                 # E1 causal promotion
# scale, learning, LLM
exp_u7_seven_arms.py --seeds 0 12 --episodes 1000   # U7/U7b/U7c/X1
exp_m1_metarl.py    --generations 40 --pop 32 --episodes 30
exp_g1_graph_analogy.py --seeds 0 12 --episodes 300
exp_l1_llm_endtoend.py --backend hf --model Qwen/Qwen2.5-7B-Instruct \
    --mode long --layouts 12          # L1/L1b (also llama3.1 via ollama)
# social ladder
exp_s1_social.py  --seeds 0 12 --episodes 300 --agents 6
exp_s2_worldclock.py --seeds 0 12 --episodes 300 --agents 6
exp_s3_admission.py  --seeds 0 12 --episodes 300 --agents 6
# integration + acceptance test
exp_r1_runtime_checklist.py                  # R1 single-run integration
exp_i1_integration.py --config songline_full --seeds 100 112 \
    --episodes 300 --agents 6        # I1 (17 configs; +--noise +--agents)
exp_ue1_utility_estimator.py --train-seeds 0 20 --test-seeds 100 110
exp_n1_noise.py   --seeds 30 --consensus 2                 # N1v2
# B1 equal-budget: run the cap at 1x, 2x, 4x the full method's footprint
exp_i1_integration.py --config raw_history --seeds 100 112 --episodes 300 \
    --agents 6 --mem-cap-kb 45 --wire-cap-kb 55    # 1x
exp_i1_integration.py --config raw_history --seeds 100 112 --episodes 300 \
    --agents 6 --mem-cap-kb 90 --wire-cap-kb 110   # 2x
exp_i1_integration.py --config raw_history --seeds 100 112 --episodes 300 \
    --agents 6 --mem-cap-kb 180 --wire-cap-kb 220  # 4x
# P1 adversarial provenance (one poisoning peer)
exp_i1_integration.py --config songline_full --seeds 100 112 --episodes 300 \
    --agents 6 --poison 1 --launder 1             # P1
exp_c1_continuous.py --config songline_safe --seeds 100 112 \
    --episodes 300 --agents 6        # C1 continuous + three-layer safety
# external substrate: the whole runtime on VMAS physics (V2.1, -53.2%)
exp_v2_vmas_runtime.py --arm songline_safe --seeds 0 12 --agents 4 \
    --max-steps 600 --out tmp/v2_full
exp_v2_vmas_runtime.py --arm independent   --seeds 0 12 --agents 4 \
    --max-steps 600 --out tmp/v2_full
# main equal-budget benchmark, 30-seed rerun (BU.1/BU.2), run ON SPHINX:
#   bash cluster/submit_bench30.sh          # 6 policies x 30 seeds (100-129)
#   python scripts/aggregate_bench30.py tmp/cluster/song_grammar/bench30
```

Registered predictions, per-run rows and verdicts sit beside each output; failed first registrations (U1 v1, U2 v1, N1 v1, C1-safe depth-3) are retained with revision notes. Cluster launcher: `cluster/submit_song_grammar.sh`. Full method freeze, claim–evidence matrix and per-verdict source map: `docs/SONGLINES_V1_FREEZE.md`, `docs/CLAIM_EVIDENCE_MATRIX.md`, `docs/SERIES_VERDICTS.md`. Design documents: `docs/FRONTIER_{SONG_GRAMMAR,UCSM}_*.md`, `docs/MEMORY_REFRAMES_2026-07-26.md`.

Tables, figures, environment, and expected cost. All tables and figures in this paper regenerate from the raw run outputs with `python scripts/aggregate_song_grammar.py` (long CSV) and `scripts/make_figures.py`; `scripts/reproduce_core.sh` chains the central deterministic blocks end to end. The pinned environment is `environment.yml` (local: Python 3.9, numpy 2.0.2; cluster base: Python 3.13, numpy 2.5.1, `vmas` 1.5.2); the GPU LLM arms use the cluster `gpu-env` (torch cu124). Expected cost of the heavy blocks: U7 at 1000 episodes ≈ 24–48 h per seed-shard on a CPU node; the L1/ALFWorld HuggingFace arms run on a 24 GB SLURM GPU node (7B fp16); the VMAS external runtime is CPU, ≈4 s for all 12 seeds; the 30-seed benchmark is 18 CPU shards,

≈ 1.5 h wall-clock; the local grid sweeps and Ollama studies are minutes-to-hours on the workstation. Test seeds ($100+$) were frozen before any tuning.

E Asset and license note

The experiments use MiniGrid and MiniWorld (Chevalier-Boisvert et al., 2023). Locally implemented benchmark and memory code will be released under the MIT License as part of the artifact package.

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